Master's Thesis

Instrument-Agnostic Iso-Mission-Time Noise Budgets for LIFE Using a Null-Order Proxy

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Abstract

The Large Interferometer For Exoplanets (LIFE) aims to detect and characterize nearby terrestrial exoplanets in the mid-infrared. Its scientific return depends on how much additional instrumental random noise can be tolerated without making the required observing program too long. This thesis develops an Agnostic Mission Simulator (AMS) and a trade-space explorer that convert a wavelength-dependent aggregate random-noise allowance into a 90th-percentile mission time. The source calculation was reformulated analytically for circularly symmetric astrophysical backgrounds, eliminating the former spatial pixel grid, correcting a normalization error of about 27% in the inherited local-zodiacal term, and permitting accurate wavelength-bin quadrature. That correction alters the derived noise allowance by factors of 1.5 to 11, an amplification of the underlying background error that follows from defining a requirement where the mission-completion curve is steep. The AMS reproduces the shared LIFEsim source-model SNRs to floating-point precision.

Final calculations compare three additive budget shapes—flat, short-weighted, and long-weighted—for the Hab2Max and Hab2Min catalogs at limiting magnitude 7, field of regard 65°, and 12 h slew time. For the second-order double-Bracewell reference, the short-weighted ramp admits the largest endpoint, consistent with geometric stellar leakage dominating the short-wavelength variance. A fourth-order null suppresses that leakage by about four orders of magnitude and the ordering reverses toward long wavelengths, where the zodiacal backgrounds still conceal added noise. The reversal holds in both catalogs, at every mission-time target, and for a realizable six-aperture combiner as well as for an idealized response.

The null-order comparison is made between architectures rather than between idealized responses. The transmission is derived from aperture positions and a beam-combiner matrix, so any physically realizable nulling design can be evaluated, and because the rotation average of a single-baseline geometry reduces to one dimension it is evaluated at the cost of the closed-form reference. Two consequences follow. A four-aperture array admits only one fourth-order output, and chopping requires two, so a fourth-order null is incompatible with LIFE's chopping scheme at four apertures; six are the minimum. At equal collecting area a six-aperture fourth-order design completes the programme about six percent sooner than the reference while tolerating a comparable flat allowance, although it is the weaker instrument for most of the catalog and gains only on the best targets, which are the ones the experiment observes. These values remain iso-mission-time amplitudes within tested budget families, not wavelength-by-wavelength subsystem requirements.

Contents

Chapter 1

1

Introduction

Why LIFE matters and how a science goal becomes an instrument requirement

Human curiosity extends far beyond Earth. We seek to understand how life began and whether Earth is its only home in the Galaxy. The Large Interferometer For Exoplanets (LIFE) turns that broad question into a concrete observational program: detecting nearby temperate terrestrial planets in the mid-infrared, measuring their thermal spectra, and using those measurements to constrain planet populations, atmospheric composition, and possible habitability [1, 3].

For me, this scientific promise makes the practical definition of LIFE especially consequential. A mission can deliver those measurements only if its instrument can be built and operated within credible performance limits. This project therefore asks how much aggregate instrumental random noise LIFE can tolerate, and where in wavelength that noise is least damaging, before the required observing program becomes too long. Answering that question helps define instrument boundaries, test the feasibility of the approach, and increase confidence in both the mission concept and the conclusions drawn from its measurements.

The reference LIFE studies used here establish the astrophysical source model, signal extraction, and expected detection and characterization performance for a double-Bracewell interferometer [1, 2, 7]. An aggregate instrumental random-noise term is not itself new, and Section 1.2 records what this work inherits. What the reference studies do not provide is its inversion: the wavelength-dependent *iso-mission-time* boundary, that is, the set of noise-budget amplitudes that all lead to the same required mission time, for a mechanism-independent aggregate random-noise term. This thesis fills that narrower gap with two linked tools. The *Agnostic Mission Simulator* (AMS) maps a specified noise budget and operating point to mission time; the *Trade Space Explorer* (TSE) inverts that mapping to find the largest tolerated budget amplitude. A controlled change from a second- to a fourth-order transmission proxy then moves the spectral region that tolerates the most added noise from short to long wavelengths. The direction of that change follows from the known suppression of geometric stellar leakage at a deeper null; what the completed exploration establishes is that the reversal survives the full nonlinear scheduling and percentile chain, and that it holds under equal integrated added noise rather than only under equal ramp endpoints.

This work keeps the double Bracewell as its reference geometry. It is mechanism-agnostic only for an aggregate random-noise allowance: individual independent random terms are represented by their common equivalent rate. The result is therefore a bridge quantity. It translates a science-level completion time into an allowed single-output random count rate at a defined *pre-efficiency reference plane*. That plane is the point in the signal chain at which photon rates have been multiplied by the collecting area but not yet by the optical throughput or the detector quantum efficiency, so a rate quoted there is independent of how those two factors are eventually apportioned; Section 2.6 fixes it precisely. A later engineering allocation can map detector, optical, thermal, and control-noise terms to that plane and test their combined variance against the allowance. This thesis does not perform that subsystem allocation and cannot decide which hardware implementation is easiest. Architecture dependence is parameterized only through the explicitly limited null-order proxy, not through an arbitrary-architecture simulation.

1.1 Research Questions

This thesis addresses the following central question.

Using an efficient architecture-agnostic simulator for arbitrary physics-conforming nulling and noise models, can instrument-agnostic requirements be determined on the aggregate random-noise error budget of the LIFE mission such that it completes Experiment 1—the inherited detection-and-characterization program for 50 qualifying catalog planets—within a reasonable mission time, under pessimistic assumptions on the physical limits?

The question names a means as well as an end. The end is a requirement: one aggregate allowance on independent random noise, referred to a defined plane and expressed so that a later subsystem allocation can be tested against it. The means is a simulator that is agnostic in two distinct senses, one about the noise and one about the instrument. It must represent the noise without committing to a physical mechanism, and it must represent the interferometer without committing to a particular beam combiner, since a requirement that holds only for one architecture is not a requirement on the mission. That the simulator must also be efficient is not a convenience: the inversion evaluates the same mission many times, and a requirement that cannot be recomputed cannot be revised.

The question is therefore structured into the following sub-questions:

- *What is the dimension of the simulator's trade space, and can it be reduced without compromising fidelity?*
- *Which source-model computations can be reused when only global filters or budget amplitudes change, and what does that reuse buy?*
- *Can the response of an arbitrary, physically realizable nulling architecture be evaluated at the cost of the closed-form reference, rather than by a per-target spatial grid?*
- *Can the maximum tolerated amplitude be located for flat and linear-ramp random-noise budgets at a fixed, deliberately pessimistic operating point?*
- *Does the resulting requirement depend on the nulling architecture, and if so, in what way?*

1.2 Relation to Prior Work

The tools and the source model used here are inherited, so it is worth stating precisely what is taken over and what is added. LIFE paper II supplies the double-Bracewell response, the signal-extraction formalism, and the LIFEsim source model on which the simulator in Section 3.6 is built, and its appendix constrains instrumental perturbations mechanism by mechanism [2].

The direct predecessor is Dannert's doctoral thesis [7]. It already introduces an aggregate instrumental photon-noise term that collects all additional photon-noise contributions, defines a fundamental-noise-limited criterion comparing that term against the astrophysical floor, and studies the resulting effect on detection yield. The aggregate random-noise quantity used here is therefore not a new object, and the present work does not claim it as one. What is new is the direction in which the mapping is used. Where the predecessor propagates an assumed instrumental noise level forward to a yield, this thesis fixes the science outcome—Experiment 1 completed within a stated mission-time target—and inverts for the largest random-noise amplitude compatible with it, resolved in wavelength across a tested family of budget shapes.

Two earlier master's theses occupy neighbouring ground, one from this group and one from a different institution. Norbruis [13], at Delft University of Technology, built a LIFEsim-derived performance model, used P-Pop populations as a test set, and ran grid and multi-objective optimisation over array geometry, aperture area, and instrument temperature to produce Pareto fronts against a detection figure of merit. That work optimises the design for yield; the present work holds the design fixed and solves for a tolerated noise allowance at fixed mission time. Binkert [14] addresses signal extraction and molecular-feature detection during the search phase, a different layer of the same mission and without overlap with the question asked here.

Against that background, four things in this thesis are new. The first is the iso-mission-time inversion itself. The second is the closed-form treatment of the circularly symmetric backgrounds, which removes image size as a production-science parameter (Section 4.2). The third is a transmission model that takes an interferometer's aperture positions and beam-combiner matrix rather than a fixed analytic response, so that any physically realizable nulling architecture can be evaluated, and evaluated at the cost of the closed-form reference rather than at the cost of a spatial grid (Section 3.3). The fourth is the finding that the spectral preference between budget shapes inverts with null order, survives the full scheduling and percentile chain rather than following only from the underlying leakage scaling, and survives the substitution of a realizable combiner for an idealized response.

The third of these deserves a remark, because it is what allows the central question to be answered as posed rather than in a weakened form. A requirement derived through a response written in closed form for one array is a requirement about that array. Only a model that accepts an arbitrary combiner can distinguish the part of the answer that belongs to the mission from the part that belongs to the instrument, and Section 5.7 shows that the two differ substantially.

The remainder follows the causal chain from mission goal to engineering requirement. Section 2 establishes the experiment, target populations, measurement, SNR, and error budget. Section 3 builds the simulator and the transmission model that lets it represent an arbitrary architecture. Section 4 reduces the raw trade space and derives the analytic background response. Section 5 then presents the completed exploration, first for the reference double-Bracewell array and then for a realizable fourth-order design. The discussion and conclusion state what the resulting allowances do and do not imply for LIFE.

Chapter 2

2

Background and Procedures

From a synthetic planetary population to a measurable mission signal

This project is conducted within the scope of the LIFE space mission. To connect its scientific objective to an engineering requirement, we first need to establish what the experiment asks the instrument to observe and how those observations become a mission-time calculation. LIFE is a proposed space-based mid-infrared nulling interferometer for detecting and characterizing nearby temperate terrestrial planets [1, 2]. Its current concept separates light collection from beam combination: four free-flying collector spacecraft feed a central beam-combiner spacecraft while the formation rotates about the line of sight. The long effective baseline supplies angular resolution that a single space telescope of comparable total collecting area could not provide.

The scientific reason for observing in the mid-infrared is twofold. A temperate planet emits much of its thermal radiation in this range, reducing the star–planet contrast relative to visible reflected light, and several atmospheric molecules have diagnostically useful absorption bands there. Nulling interferometry does not form a conventional resolved image of every system. Instead, it suppresses the on-axis star, modulates off-axis emission through rotation, and measures a wavelength-dependent differential signal. Detecting that signal constrains planet populations; accumulating a spectrum can constrain atmospheric composition. The latter remains an inference problem rather than an automatic identification of life [1, 3].

This thesis retains the four-aperture double-Bracewell response used by LIFEsim [2]. Two aperture pairs first produce destructive and constructive outputs. The destructive outputs are recombined with a relative phase shift to produce two chopped dark outputs whose difference changes sign for an off-axis source. The reference response has a second-order intensity null: close to the optical axis, transmitted stellar intensity grows quadratically with angular offset. Figure 2.1 shows this optical logic.

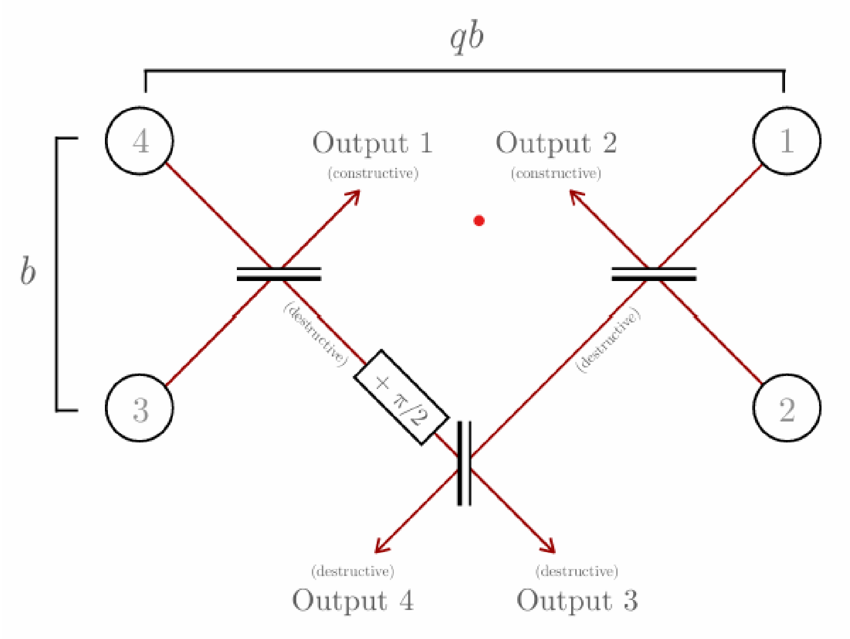


Figure 2.1: Double-Bracewell interferometer layout reproduced from Dannert et al. [2]. Light from apertures 4 and 3, and from 1 and 2, is combined to form destructive and constructive outputs. The destructive outputs are recombined with a relative phase shift of $\pi/2$ to form two chopped dark outputs with a second-order null.

2.1 The Experiment and Assumptions

This work adopts the legacy LIFEsim/AMS configuration called *Experiment 1*: the mission must detect and characterize 50 catalog planets with radii between 0.5 and 1.5 Earth radii, host-star temperatures between 4370 and 7310 K, and a positive catalog habitable-zone flag. The thresholds and instrument settings used throughout the completed data collection are summarized in Table 2.1. They define this study's controlled comparison; they should not be read as a claim that every value is the latest top-level LIFE mission requirement.

Quantity	Adopted value	Role in the calculation
Qualifying sample	50 planets	Experiment-completion target
Planet radius	0.5–1.5 $R_\oplus$	Catalog eligibility filter
Host temperature	4370–7310 K	Catalog eligibility filter
Habitable zone	Catalog flag required	Catalog eligibility filter
Detection threshold	SNR 7	Common detection campaign
Characterization threshold	SNR 44.1	Broadband characterization proxy
Orbit epochs	5	Detection plus four follow-up visits
Collector diameter	4 m each	Photon collecting area
Optical throughput	0.15	Pre-detector transmission
Quantum efficiency	0.60	Photon-to-count conversion
Total efficiency	0.09	Product of throughput and quantum efficiency
Spectral band and resolution	4–18.5 μm, $R = 20$	31 finite wavelength bins
Baseline ratio and bounds	$r = 6$, 10–100 m	Imaging/nulling geometry and clipping

Table 2.1: Fixed LIFEsim/AMS settings used for all final simulations. SNR 7 follows the LIFE yield-study detection convention [4]. The characterization value 44.1 is the integrated broadband equivalent of a legacy per-channel requirement of SNR 10 at 11.2 μm and $R = 50$ [7].

The characterization threshold deserves particular care. It is not a universal statement that every molecular feature is measurable at bulk SNR 44.1. It is a retained broadband proxy that permits a consistent mission-time comparison with the inherited simulation framework [7]. Similarly, the fixed 4 m collectors and 9% total efficiency set photon rates but are not optimized in this thesis. The output is conditional on all these assumptions.

These assumptions specify the required scientific outcome, but not the individual planets that must supply it. We therefore next define the simulated target populations on which the mission-time calculation is based.

2.2 Target Catalog Selection

The LIFE target catalogs are created from a common scaffold of 4505 nearby stars, including Sun-like and M-dwarf hosts out to roughly 50 pc. This scaffold was inherited together with the catalogs; its entries carry TESS Input Catalog identifiers, and the star count and distance limit quoted here were verified directly from the catalog files rather than taken from a published target list. It is deliberately broader than the stellar sample of LIFE paper VI, which is restricted to single and wide binary main-sequence stars within 20 pc [4]. The $\eta_\oplus$ values adopted in Table 2.2 were averaged over that smaller sample, so the occurrence normalization and the stellar scaffold used here are not drawn from the same stellar population. This is acceptable for the present purpose, because the comparison throughout is between operating points and budget shapes evaluated on one fixed scaffold, but it means the absolute planet counts in Table 2.3 should not be read against the yields reported in LIFE paper VI. Since planets are not known around most of them, the population must be synthesized. P-Pop draws occurrence and orbital properties constrained by Kepler statistics and places a hypothetical planetary system around every stellar target [5]. One complete draw over the stellar scaffold is called a *universe*. The 100 universes in a catalog are therefore alternative population realizations, not 100 independent observations of the real Galaxy.

We use two catalogs, *Hab2Min* and *Hab2Max*, with 100 universes each. They implement the lower and upper "hab2 stars" extrapolation scenarios of the Bryson et al. occurrence model as adopted in LIFE paper VI [6, 4]. The quantity $\eta_\oplus$ is the average number of exo-Earth candidates per Sun-like star, not the fraction of stars hosting at least one such planet. Hab2Min represents the less favorable and Hab2Max the more favorable occurrence scenario; their difference is driven chiefly by the extrapolation beyond 500 d, where Kepler completeness no longer directly constrains the occurrence surface. Both retain the same stars, so comparing them changes the hypothetical planets rather than the accessible stellar neighborhood. M dwarfs remain in the source catalogs but fail the Experiment 1 host-temperature cut and are not qualifying targets.

Catalog	$\eta_\oplus$
Hab2Min	$0.19^{+0.19}_{-0.10}$
Hab2Max	$0.34^{+0.38}_{-0.19}$

Table 2.2: Mean numbers of exo-Earth candidates per Sun-like star used to define the two occurrence scenarios [6, 4]. The asymmetric bounds are the reported 1σ occurrence-model uncertainties and are dominated by occurrence-rate and long-period extrapolation uncertainty.

The catalog ensemble has two roles. It exposes the scheduler to different possible planet

Catalog	Universes	Planet rows	Stars	Eligible planet rows	Eligible planets/universe
Hab2Min	100	486 244	4505	205 715	1958–2154
Hab2Max	100	729 925	4505	318 654	3049–3331

Table 2.3: Inventory measured from the exact input files used in this work after loading them through the same P-Pop interface as the AMS. "Eligible" applies only the Experiment 1 radius, host-temperature, and habitable-zone filters; visibility, SNR, and scheduling cuts are applied later. The median eligible counts are 2058.5 for Hab2Min and 3194 for Hab2Max. LIFE paper VI uses a narrower exo-Earth-candidate subset, $0.8a_p^{-0.5}R_\oplus \leq R_p \leq 1.4R_\oplus$, for its occurrence analysis; that definition does not replace the broader Experiment 1 radius filter used here [4].

populations, and it supplies the distribution from which the 90th-percentile mission time is taken. Catalog eligibility alone does not imply detectability: transmission, noise, magnitude, field of regard, and the finite observing campaign all act later. This distinction explains why target selection precedes signal collection in the narrative.

The catalog fixes the systems and planets to be observed; the next step is to determine how light from each hypothetical planet is separated from its much brighter surroundings.

2.3 Signal Collection: Transmission Map, Chopping, and Modulation

Before quantifying the SNR, we must establish how the interferometer separates an off-axis planet from bright backgrounds. A transmission map $T(\alpha, \beta)$ gives the fraction of incident flux reaching one output at sky offset (α, β). Following the notation of LIFE paper II, the nulling baseline b sets the broad null envelope, while the imaging baseline rb sets the fine chopping and modulation fringes; the baseline ratio is $r = 6$ in the nominal settings. For each star, b is selected using the habitable-zone-centre prescription and clipped to the instrument bounds. A common field-of-view taper attenuates off-axis light. The star lies on the central null, whereas an off-axis planet samples the surrounding fringes [2].

The two destructive outputs are chopped by subtraction. Their symmetric mean backgrounds cancel in the difference, but their independent photon-counting variances remain and add. A planet generally contributes to both outputs, but unequally, and therefore survives the subtraction (Fig. 2.2a, right).

The final ingredient is *modulation* through array rotation. Over an observation the entire array is slowly rotated about the line of sight. A planet at a fixed angular separation is thereby swept through the alternating positive and negative lobes of the chopped map, so that its contribution to the differential output oscillates in time, tracing out a characteristic *transmission curve* as a function of rotation angle (Fig. 2.2b). The symmetric backgrounds, being nearly axisymmetric, stay approximately constant under this rotation: Lay reports that the stellar and local-zodiacal geometric leakage rates are nominally independent of the array rotation angle and therefore appear at zero frequency, and Dannert et al. state correspondingly that for rotationally symmetric sources the detected signal does not depend on the rotation angle of the array [9, 2]. The planetary signal is therefore encoded as a time-modulated component riding on an essentially static background, which allows it to be isolated in subsequent analysis [10, 2]. This is the origin of the temporal average $\langle\cdot\rangle$ over one full array rotation that appears throughout the SNR derivation below.

The calculation does not simulate a time-series estimator sample by sample. It uses the root-mean-square response over a complete, uniformly sampled rotation. For a chopped planet transmission $m(\theta) = T_3(\theta) - T_4(\theta)$, the effective signal coefficient is therefore

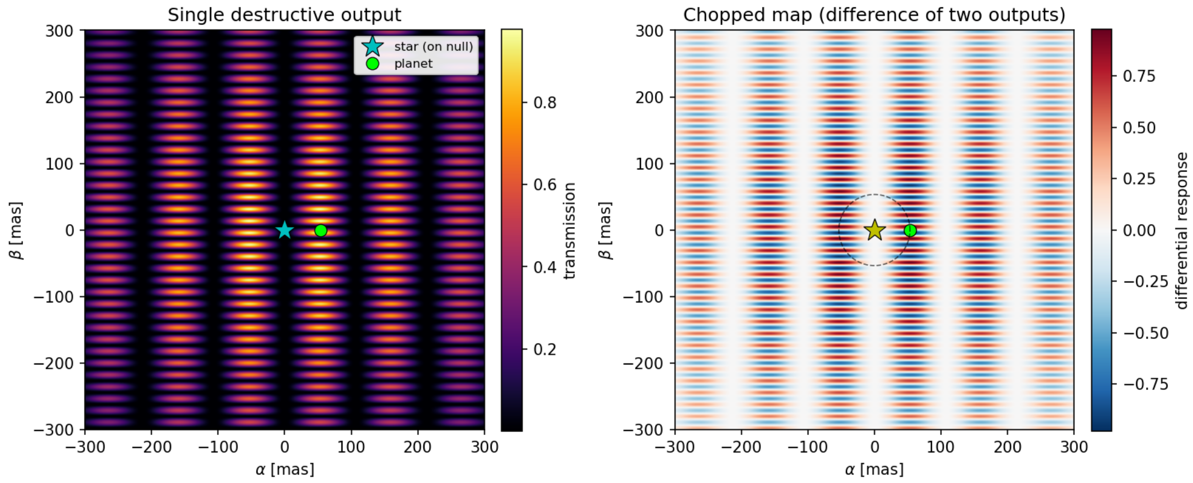
$$s = \left[\frac{1}{2\pi}\int_0^{2\pi} m^2(\theta)\,\mathrm{d}\theta\right]^{1/2}. \tag{2.1}$$

This is the matched-modulation amplitude for white, rotation-independent variance under the adopted normalization. A constant symmetric brightness cancels from the chopped mean, but it still sends photons to each output and therefore remains in the variance. Figure 2.2 shows the single-output and chopped maps over which these coefficients are averaged.

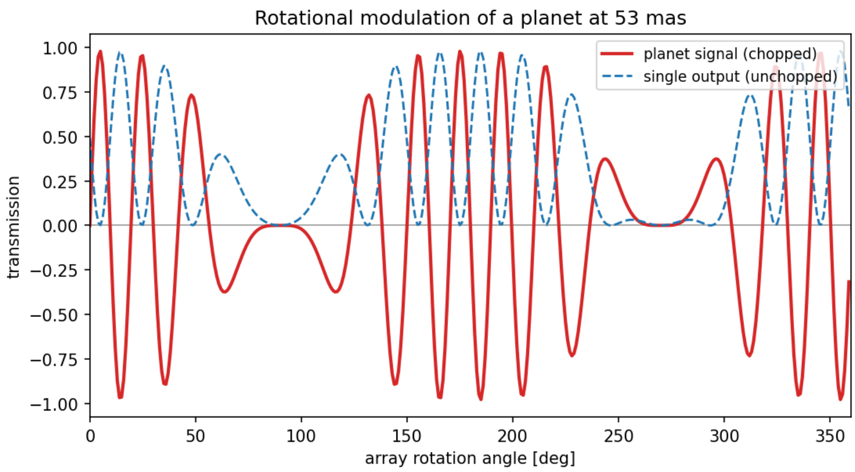
Two consequences of this scheme carry directly into the SNR. First, the planetary *signal* is collected through the chopped map, whereas the planetary *photon noise*, which is insensitive to the sign of the modulation, is collected through a single, unchopped output; the two therefore enter with different transmission weights. Second, because the map is integrated over a full rotation, the effective response becomes radially symmetric and depends only on the angular separation of the planet and on the nulling baseline. This rotation-averaged response, further attenuated by a field-of-view taper that models the loss of off-axis coupling into the instrument's optical fibers, is the *transmission efficiency* introduced in Section 2.6. Having established what reaches the detector and how it is weighted, we can now quantify whether the planet can be distinguished from noise.

2.4 The Signal-To-Noise Ratio (SNR)

The SNR is the expected modulated count signal divided by the standard deviation of the associated random count fluctuation. The implementation evaluates finite wavelength



(a) Transmission maps of the double Bracewell array.



(b) Rotational modulation of the marked planet.

Figure 2.2: Signal collection in the double-Bracewell reference geometry at 10.4 μm, $b = 20\,\text{m}$ and nominal baseline ratio $r = 6$, zoomed to the central $\pm300\,\text{mas}$. (a) A single destructive-output map and its chopped difference. Symmetric mean contributions cancel in the difference, whereas an off-axis planet generally has unequal transmission in the two outputs. (b) Rotation modulates the chopped planet signal; the unchopped single-output response used for its photon-noise term has a different weighting.

bins. Its reference planes are easy to confuse because planet flux is naturally specified at the observer, whereas extended backgrounds have already been integrated over collecting area. Table 2.4 makes the convention explicit.

For bin i, the expected chopped planet counts follow from its incident flux, collecting area, efficiency, time, and RMS modulation response. Each destructive output is a Poisson stream. If the two output counts have means μ_3 and μ_4, then

$$\mathrm{Var}(C_3 - C_4) = \mathrm{Var}(C_3) + \mathrm{Var}(C_4) = \mu_3 + \mu_4 \tag{2.2}$$

Symbol	Meaning at the SNR interface	Units
$F_{p,i}$	Planet photon flux at the collectors, integrated over bin i	$\mathrm{ph\,s^{-1}\,m^{-2}}$
A	Total effective collecting area in the source convention	$\mathrm{m^2}$
η	Optical throughput times quantum efficiency	1
s_i	RMS chopped planet-response coefficient	1
n_i	RMS single-output planet-noise coefficient	1
B_i	Single-output astrophysical rate after area, before η	$\mathrm{ph\,s^{-1}}$
$N_{I,i}$	Added single-output rate after area, before η	$\mathrm{ph\,s^{-1}}$
t	On-source integration time	s

Table 2.4: SNR symbols and reference-plane convention. The reported spectral budget $N_I(\lambda)$ is converted to a bin rate by $N_{I,i} = N_I(\lambda_i)\Delta\lambda_i$.

when the streams are independent. Symmetric backgrounds have equal rotation-averaged rates in both outputs, so subtracting their means removes the DC signal but doubles their single-output variance. With $s_i = \sqrt{\langle (T_3 - T_4)^2 \rangle}$ and $n_i = \sqrt{\langle T_4^2 \rangle}$, the implemented expected signal counts and variance are

$$C_{S,i} = F_{p,i} A \eta t s_i, \tag{2.3}$$

$$V_i = 2\eta t \left[B_i + N_{I,i} + A F_{p,i} n_i \right]. \tag{2.4}$$

Hence

$$\mathrm{SNR}_i = \frac{C_{S,i}}{\sqrt{V_i}}. \tag{2.5}$$

Rates have units $\mathrm{ph\,s^{-1}}$ per bin, $C_{S,i}$ is a count, V_i a count variance, and $\sqrt{V_i}$ a count standard deviation. Chopping cancels the symmetric mean but subtracts two destructive outputs, so their independent photon-counting variances add and give the factor two. Output-swap/rotation symmetry makes the rotation-averaged single-output planet photon-noise rates equal; the same two-output variance sum therefore gives the factor two for the planet term. Assuming independent bins, the bulk SNR is

$$\mathrm{SNR}_{\mathrm{tot}} = \sqrt{\sum_{\lambda_i} \mathrm{SNR}_{\lambda_i}^2}. \tag{2.6}$$

It scales as $\sqrt{t}$; $\mathrm{SNR}_{1\mathrm{h}}$ uses $t = 3600\,\mathrm{s}$. This source-model convention follows LIFE paper II [2]. Correlated fluctuations and systematic biases are outside this photon-statistical model [9, 7]. From this point onward, SNR refers to bulk SNR unless otherwise stated.

The SNR expression identifies the quantities that limit a measurement, but the error budget can only be defined after those noise contributions have been separated by physical origin.

2.5 Non-Systematic Noise Sources

The model distinguishes noise by both physical origin and statistical treatment. Stellar leakage arises because a real star has finite angular radius: only its centre lies exactly on the null, while light from the resolved disk is transmitted. Local-zodiacal emission is thermal radiation from dust in the Solar System; exozodiacal emission is the analogous dust contribution around the target star. The adopted exozodiacal surface-brightness model follows the smooth, face-on disk prescription used in LIFEsim and is tied to established exozodiacal modeling [8, 2]. All three contribute photon shot noise even where chopping cancels their symmetric mean.

Table 2.5 collects these terms with their physical origin, the dependence retained in the model, and the boundary of that treatment.

Term	Physical origin	Modeled dependence	Treatment and boundary
Stellar leakage	Finite stellar disk transmitted around the central null	λ, stellar radius and temperature, distance, b, null order	Uniform circular disk; photon noise only
Local zodiacal	Solar-System dust foreground	λ, target ecliptic latitude, field of view	Locally uniform circular foreground; photon noise only
Exozodiacal	Dust in the target system	λ, stellar luminosity and distance, disk radius	Smooth face-on axisymmetric disk; photon noise only
Planet	Desired thermal emission	λ, radius, temperature, separation and rotation	Chopped signal plus its own photon noise
Added budget N_I	Equivalent aggregate instrument contribution	Imposed spectral family and amplitude	Independent Poisson-like variance at the pre-efficiency single-output plane
Excluded terms	Calibration bias, drift, correlated instability, confusion	Architecture and data-reduction dependent	Not represented by N_I

Table 2.5: Noise taxonomy and model boundary [2, 8]. "Photon noise only" means that a source's mean chopped signal may cancel while the associated shot-noise variance remains. Excluded systematic and instability terms require a separate architecture-dependent treatment [9, 7].

The question is how much instrumental random noise can be allowed in addition to the astrophysical backgrounds while retaining the adopted mission target. Here the *random-*

noise error budget is operationally an equivalent added single-destructive-output photon-rate spectral density $N_I(\lambda)$, in $\mathrm{ph\,s^{-1}\,\mu m^{-1}}$, referred to the same pre-efficiency plane as the astrophysical background. It represents independent Poisson-like random terms that enter as additive variance after differencing. It neither represents correlated or systematic biases nor assigns the allowance to individual subsystems. To turn that budget into an engineering question, we next express the physical contributions as parameters that can be varied or held fixed.

2.6 Parameters and Instrument Performance

A *parameter* is a value or function used to describe a candidate mission calculation. A *variable* parameter is a target-dependent physical signal or noise channel; changing it triggers SNR recomputation (for example stellar leakage or planetary transmission). An *agnostic* parameter is a global visibility filter or overhead applied without recomputing source physics (for example limiting magnitude, field of regard, or slew time).

The word *agnostic* is used in two distinct senses in this thesis, and they should not be conflated. In the title and in Section 1 it is *mechanism*-agnostic: the instrumental allowance $N_I(\lambda)$ does not commit to a physical origin for the random terms it aggregates. In the sense just defined, and throughout the description of the simulator, it is a *computational* classification: an agnostic parameter is one whose variation does not require the source physics to be recomputed. The first sense is a modelling choice about what the result means; the second is an implementation property that makes the repeated evaluations affordable. A parameter can be agnostic in the second sense while being entirely architecture-specific, as slew time is.

The ideal $N_I = 0$ case is the physical lower bound: it has maximum SNR and minimum mission time, but it is the *hardest* engineering requirement because it demands a perfectly noiseless added contribution. It is a mathematical reference origin, not an easy design. Negative budgets are outside this model. Requirements are relaxed upward from this origin until the iso-mission-time boundary is reached. If, at some wavelength, the added budget is lower than the total astrophysical noise, the calculation is *astrophysical-noise-limited* there; if it exceeds it, it is *instrumental-noise-limited*.

As an example, consider again the stellar leakage. Typically, a star emits more strongly in the shorter wavelengths, so we might add some margin to the shorter wavelengths and observe the impact on mission time. This motivates the short-weighted ramp tested in Section 5; the opposite ramp tests the zodiacal-dominated long wavelengths. Neither shape redistributes or subtracts noise elsewhere.

Planetary emission enters through the rotation-averaged transmission efficiency and a common field-of-view taper. As defined in Section 2.3, b sets the null envelope and rb the fine fringes in nominal settings; b is selected per star from the habitable-zone centre and clipped to the allowed bounds. The field-of-view taper represents common off-axis coupling loss without assuming a particular taper formula.

We can now collect the complete set of variable parameters and the physical quantities each of them depends on. Every noise contribution reaches the instrument through the transmission map, which is why each additionally carries the wavelength dependence characteristic of its blackbody source. The variable parameters are as follows:

- Stellar leakage. Depends on the wavelength, the stellar temperature, the stellar radius, and the distance to the star.
- Local-zodiacal noise. Depends on the wavelength and the ecliptic latitude of the target.
- Exo-zodiacal noise. Depends on the wavelength, the stellar luminosity, and the distance to the star.
- Planetary signal. Depends on the wavelength and, through the transmission efficiency, on the nulling baseline and the field of view taper.

This enumeration specifies the raw dimension more precisely. Four physical channels each carry a multiplicative and an additive modifier, giving up to eight free functions of their listed physical variables. Together with the three scalar agnostic parameters, the unrestricted problem is formally infinite-dimensional because each free function has infinitely many degrees of freedom. The reduced calculations reported here are much smaller: for fixed catalog, null order, operating point, and spectral family, the search is one-dimensional in amplitude A; including magnitude, field of regard, and slew time gives four scalar dimensions; allowing the two ramp endpoints to vary independently gives five. Catalog, null order, spectral family, and target mission time remain discrete scenario choices. This hierarchy motivates the reductions of Section 4.1.

In addition to the variable ones, there are parameters termed *agnostic*: global filters or overheads scanned without recomputing source physics, not quantities literally independent of every architecture. They are evaluated on simple grids. The agnostic parameters are as follows:

- Limiting magnitude. Describes the maximum K-band magnitude object that the instrument can detect.

- Field of regard. The implemented filter marks a target visible when its absolute ecliptic latitude is at most the FoR. Thus FoR 65° retains the band from −65° to +65° and excludes the polar caps; FoR $\pi/2$ retains the full sky.

- Slew time. Defines the average time it takes for the instrument to center on a new target.

Knowing the starting points and limits, we can now go back to the performance of the instrument. We employ a practical science iso-performance approach. Specifically, we declare an instrument better-performing if it can complete the scientific goals of the mission in less time. The variable and agnostic parameters together span a large space of possible choices, which we call the *trade space*. To estimate an error budget, the *Agnostic Mission Simulator* (AMS), described in Section 3.6, maps inputs to mission time. Here a set budget means a specified wavelength-dependent additive shape together with selected agnostic operating parameters.

2.7 The Trade Space

The parameters define a forward map from an instrument description to mission time, but the engineering question asks for its inverse: the largest added budget compatible with a chosen time. The raw space cannot be sampled directly because it contains free functions. Even a finite ten-dimensional surrogate, sampled at only ten values per dimension, would require 10^{10} AMS evaluations. At an optimistic one second per evaluation, that coarse grid would take approximately 317 years. This estimate is illustrative rather than a claim that the raw functional space has exactly ten dimensions. It shows why the workflow must first restrict the budget to interpretable spectral families, then exploit monotonic searches and reusable computations instead of applying a uniform grid.

The completed TSE runs invert this mapping for null orders two and four. They therefore show both how much aggregate random-noise amplitude is tolerated within each tested shape and how the modeled null order changes the spectral ordering of those allowances.

Chapter 3

3

Methods

Building a fixed-reference simulator and a controlled null-order proxy

The background has established the quantities mapped from a fixed double-Bracewell reference description to mission time. We now construct that mapping, with a mechanism-agnostic aggregate random-noise allowance and the stated null-order proxy.

3.1 Variable Parameter Treatment

A free interpolation grid would preserve maximum flexibility but create a poorly constrained functional search. Instead, the AMS modifies each modeled photon-flux density F through a multiplicative function M and an additive function A:

$$\mathrm{V}(\alpha_1, \ldots, \alpha_n) = \mathrm{M}(\alpha_1, \ldots, \alpha_n) \times \mathrm{F} + \mathrm{A}(\alpha_1, \ldots, \alpha_n), \tag{3.1}$$

where M is a multiplicative modifier and A an additive modifier. For an *extra-noise* channel, $M \geq 1$ and $A \geq 0$ preserve the astrophysical reference, with the zero-budget case $M = 1$, $A = 0$. These inequalities do not describe signal degradation: that requires a separate convention, such as $0 \leq M \leq 1$ or a negative additive term constrained to keep the signal non-negative. The condensed study reported here varies additive background noise only.

3.2 The Reference Double-Bracewell Response

The simulator remains mechanism-agnostic with respect to the aggregate random-noise allowance throughout. Its treatment of the instrument, by contrast, begins from the reference array and is then generalized. For the reference double Bracewell, the first pairwise combinations produce a $\sin^2$ nulling envelope, while the second combination produces complementary imaging fringes. Let α and β be angular sky offsets along the nulling and imaging-baseline directions. With the LIFE notation b for nulling baseline and r for the imaging-to-nulling baseline ratio [2], the phases are $\phi = \pi b\alpha/\lambda$ and $\psi = \pi r b\beta/\lambda$. The two reference destructive outputs are

$$T_{3,2} = \sin^2(\phi)\cos^2(\psi - \pi/4), \tag{3.2}$$

$$T_{4,2} = \sin^2(\phi)\cos^2(\psi + \pi/4). \tag{3.3}$$

Their difference is $T_{3,2} - T_{4,2} = \sin^2(\phi)\sin(2\psi)$, up to the chosen output-sign convention. Close to the optical axis, $\sin\phi \simeq \phi$, so the nulling envelope grows as ϕ^2 and has intensity null order two.

Written this way the response is exact, but it is exact for one array. A requirement derived through it is a statement about the double Bracewell rather than about the mission, and the central question asks for a simulator that accepts arbitrary physics-conforming nulling models. The remainder of this section replaces the closed form with a description that any nulling interferometer satisfies.

3.3 Transmission of an Arbitrary Nulling Architecture

A nulling interferometer is specified completely by where its collectors sit and how their beams are combined. Let the apertures occupy positions $\mathbf{r}_k$ in the array plane and let the beam combiner be described by a matrix U with one row per output and one column per aperture. A source at sky offset $\boldsymbol{\theta}$ imposes a phase $2\pi\,\mathbf{r}_k\cdot\boldsymbol{\theta}/\lambda$ on aperture k, so output j receives

$$T_j(\boldsymbol{\theta}) = \frac{1}{N}\left|\sum_{k=1}^{N} U_{jk}\,\exp\!\left(\frac{2\pi i\,\mathbf{r}_k\cdot\boldsymbol{\theta}}{\lambda}\right)\right|^2, \tag{3.4}$$

with N the number of apertures. Everything the simulator needs follows from Eq. 3.4: the null order, the chopping behaviour, the rotation-averaged background response, and the throughput. Nothing is assumed about the architecture beyond the array being lossless, which makes the rows of U orthonormal.

The factor $1/N$ fixes a convention that matters as soon as designs with different aperture counts are compared. With unit amplitude at each aperture the array collects N units of intensity and a lossless combiner conserves it, so dividing by N makes the outputs sum to unity and a response is then directly the fraction of collected light reaching that output. Throughput therefore ceases to be a separate quantity requiring its own bookkeeping: it is the number of outputs used for science divided by N. This is worth stating explicitly, because holding throughput fixed while changing the response is exactly the error an idealized shape model invites.

Equation 3.4 reproduces the closed form above when U is the double Bracewell. Expressed as four apertures on a rectangle with the nulling baseline along α and the imaging baseline along β, and with the two Bracewell nulls recombined at relative phase $\pi/2$, it agrees with $T_{3,2}$ and $T_{4,2}$ to a relative 3×10^{-14}, and an independent power-law fit to its near-axis response returns a null order of 2.0000 without being told what to expect.

3.4 Architectures Considered

Three designs are used in this work, summarized in Table 3.1. The first is LIFE's reference array. The second and third achieve a fourth-order null, and comparing them establishes a constraint that no idealized response can express.

Architecture	Apertures	Null order	Deep outputs	Chopping	Light in pair
Double Bracewell (reference)	4	2	3	yes	50 %
Angel Cross	4	4	1	**no**	25 %
Double triple nuller	6	4	2	yes	33 %

Table 3.1: Architectures evaluated in this work. Deep outputs and the fraction of collected light follow from the row structure of an orthonormal combiner matrix rather than from any fitted quantity; the 50 % and 25 % figures reproduce those quoted for Darwin and the Angel Cross by Guyon et al. [11].

The Angel Cross places two crossed Bracewell pairs on a square. Its four orthogonal outputs comprise one bright output, two second-order outputs, and a single output whose amplitude vanishes to second order and whose intensity therefore vanishes to fourth. That single deep output is the constraint: chopping requires two dark outputs whose difference changes sign for an off-axis source, and a four-aperture fourth-order array has only one. **A fourth-order null at four apertures is therefore incompatible with the chopping scheme LIFE relies on**, not as a matter of engineering difficulty but as a consequence of counting orthogonal outputs.

Six apertures are the minimum that restores it. The double triple nuller is the direct analogue of LIFE's own architecture with each two-aperture Bracewell pair replaced by a three-aperture nuller of amplitudes $1 : -2 : 1$. A single such triple has amplitude proportional to $2(\cos\psi - 1)$, second order in the offset, so its intensity vanishes to fourth order; this is the one-dimensional $2(N-1)$ rule for $N = 3$ [11]. Two triples separated along the imaging baseline and recombined at relative phase $\pi/2$, exactly as the two Bracewell pairs are in the reference design, yield two fourth-order outputs of opposite modulation sign, and chopping is available again.

Because the achievable null order is primarily a function of aperture number, a deeper null is bought with collectors rather than with a cleverer combiner at fixed N [11]. Kernel nullers formulated for an arbitrary number of apertures [12] are the natural family in which further designs would be expressed, and Eq. 3.4 accepts them unchanged.

3.5 Why an Arbitrary Architecture Remains Affordable

Evaluating Eq. 3.4 on a spatial grid for every target and every wavelength would be far more expensive than the closed form it replaces, and would forfeit the reductions of Section 4. It is not necessary. The array rotates during an observation, and for a circularly symmetric source that is equivalent to averaging the response over the azimuth of the source. If the geometry scales with a single baseline, so that $\mathbf{r}_k = b\,\mathbf{u}_k$ for fixed $\mathbf{u}_k$, the phase in Eq. 3.4 depends on the target only through

$$x = \frac{\pi b\,\theta}{\lambda}, \tag{3.5}$$

the same dimensionless separation the reference response depends on. The rotation-averaged response of any such architecture is therefore a one-dimensional function of x, and can be tabulated once per design and interpolated thereafter.

The consequence is that the cost of an arbitrary combiner is the cost of the reference. Measured on the same catalog and hardware, a full evaluation takes 14.3 s with a general combiner against 14.7 s with the closed form, and the tabulated background response agrees with the analytic Bessel reduction of Appendix A to a relative 6×10^{-12}. The architecture-agnostic model is thus not a slower fallback used where the closed form is unavailable; it is the same calculation, obtained numerically rather than by hand.

3.6 The Agnostic Mission Simulator (AMS)

The AMS maps one parameter set to mission time in two phases. The *simulation phase* computes or reuses an unfiltered one-hour SNR for every catalog planet, then applies the global visibility filters. The *distribution phase* schedules the filtered targets and returns the required percentile mission time.

3.6.1 The Simulation Phase

Figure 3.1 separates the expensive *SNR flow* from the inexpensive *filter flow*. The SNR flow fills a persistent array with one-hour SNRs for every target and reruns only when a physics-governing quantity changes. The filter flow always returns a copy in which targets failing the magnitude or field-of-regard cut have SNR zero; it never overwrites the unfiltered cache.

When the SNR flow runs, it iterates over the complete catalog. Within that recomputation, stellar leakage, local-zodiacal noise, and the unit-exozodi response are reused among planets sharing one host; planet transmission remains target-specific. Magnitude, field of regard, and slew-time changes reuse the unfiltered SNR array and rerun only filtering and time distribution. By contrast, any change to budget amplitude or spectral shape changes the bin variance, invalidates the SNR array, and requires catalog-wide SNR recomputation.

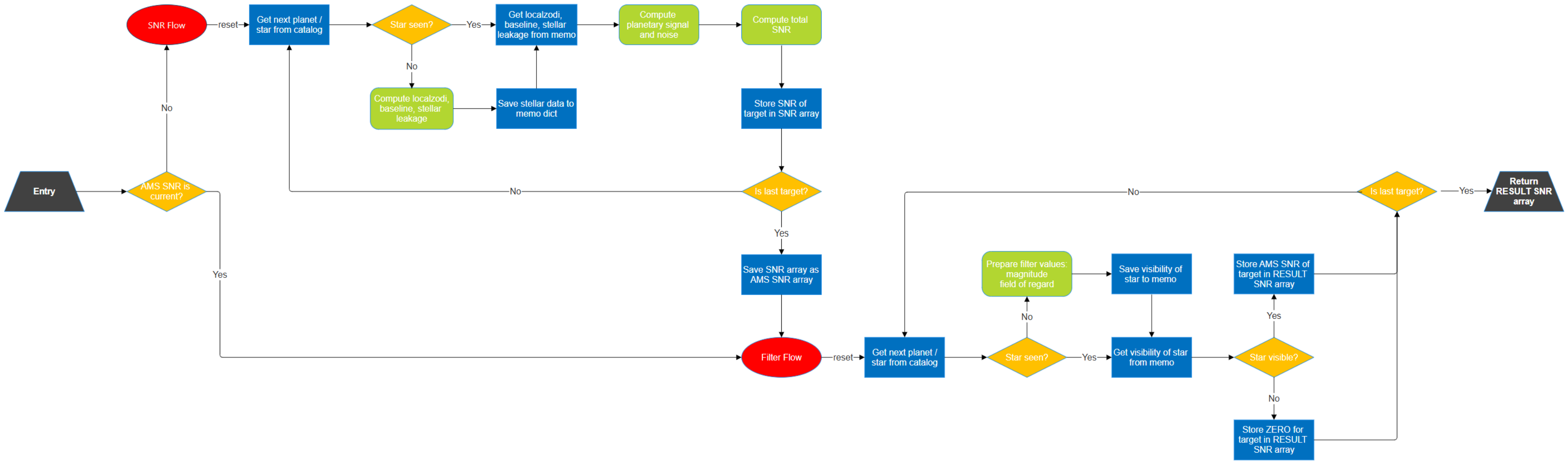


Figure 3.1: AMS simulation phase. Gray trapezoids mark entry or exit, red tiles denote flows, yellow diamonds decisions, green boxes calculations, and blue rectangles memory operations. Physics or budget changes trigger the SNR flow and replace the unfiltered cache. The filter flow always copies that cache and sets targets outside the magnitude or field-of-regard limits to zero; it never destroys unfiltered values.

3.6.2 The Distribution Phase

We now have a full set of filtered SNRs for every target in the input planet catalog. The only remaining step is to use LIFEsim to distribute the available observation time across the targets and then find the required mission time.

The distribution rests on photon-limited SNR accumulation as the square root of integration time. The SNR flow yields $\mathrm{SNR}_{1\mathrm{h}}$ for each target, so

$$t_{\mathrm{int}}(q) = \left(\frac{q}{\mathrm{SNR}_{1\mathrm{h}}}\right)^2 \quad [\mathrm{h}] \tag{3.6}$$

for a fresh integration to requested bulk SNR q. The implemented follow-up calculation does not subtract SNR already accumulated during detection. For a selected planet, with slew time t_{slew}, it charges

$$t_{\mathrm{orbit}} = (n_{\mathrm{orbits}} - 1)\left[t_{\mathrm{int}}(7) + t_{\mathrm{slew}}\right], \tag{3.7}$$

$$t_{\mathrm{char}} = t_{\mathrm{int}}(44.1) + t_{\mathrm{slew}}. \tag{3.8}$$

This convention treats the four follow-up orbit epochs and the characterization as separate observations. It is intentionally reported as implemented; alternative reuse of detection photons would reduce the follow-up cost.

The complete scheduler is:

1. Optimize one common detection sequence over all 100 synthetic universes.
2. Stop when the inherited strict completion checks report more than 50 qualifying detections in more than 90% of universes. With 100 universes, this means at least 51 planets in at least 91 universes.
3. Assign the resulting shared detection-campaign time to every represented universe.
4. In each universe, retain eligible planets that were detected, rank them by their one-hour SNR at maximum separation, and select the best 50.
5. Add four orbit follow-ups and one characterization observation per selected planet using the equations above.
6. Return the 90th percentile of the resulting per-universe total times.

The AMS is consequently a perfect-information ensemble yield model, not an adaptive discovery scheduler. In experiment-limited mode, the configured 0.8 observing efficiency neither caps nor rescales the returned time. Here "completion of Experiment 1" always inherits the strict implementation above: at least 51 detections in at least 91 universes, with empty universes omitted from the time sheet. The strict inequalities tend to raise mission time, whereas omission can bias the reported distribution low. Their net effect is not known from the retained outputs.

Two consequences follow for how every time reported in this thesis should be read. First, the returned quantity is the time the scheduler charges: the sum of on-source integration and slew, accumulated over the shared detection campaign, the four orbit follow-ups, and the characterization observation. It is not a calendar duration, because no duty cycle, downlink, calibration, or safe-mode overhead is applied to it. Second, the observing efficiency of 0.8 configured in Table 2.1 would, if applied as a duty cycle, map a charged time T to an elapsed duration $T/0.8$; the 5.5-year target used throughout would then correspond to roughly 6.9 years of elapsed mission. Every target, contour, and budget amplitude reported here is stated in charged time, so comparisons among them are internally consistent. They should not be read directly against a mission calendar, and the allowances in Section 5 are correspondingly optimistic if the reader interprets the target as elapsed time.

The returned percentile is fixed at 90%. If 100 universe times are retained, a distribution-free binomial calculation gives approximately 95.6% coverage for the population 90th percentile between the 84th and 96th order statistics. Fewer retained universes require different ranks. The saved final products contain contour endpoints but not every per-universe time vector, so this rank interval cannot be converted into a numerical uncertainty in years or budget amplitude after the fact. Future production runs should retain those vectors and report a corresponding interval alongside every result.

3.6.3 Validation of the SNR Flow

The physics and numerical rewrite were checked at several distinct boundaries; Table 3.2 states what each check can and cannot establish.

Comparison	Result	Interpretation
Shared-route SNR comparison (two thousand sampled targets)	Maximum relative difference 4.2×10^{-16}	Confirms wiring and vectorization against shared source modules
8-point versus 64-point inverse-wavelength quadrature, $R = 5, 10, 20, 50$	Approximately machine precision for sampled stellar and exozodiacal terms	Establishes convergence of within-bin integration for the retained smooth spectra
Analytic azimuthal response versus 2×10^6 angle samples, orders 2, 4, 6	Relative agreement 10^{-16} for $x \gtrsim 1$, rising to 10^{-12} at $x = 0.2$ and order six	Checks the general even-order Bessel reduction; the larger residual is cancellation in the coefficient sum where the response is itself of order 10^{-5}
Uniform-disk average versus dense two-dimensional grid	Agreement at approximately 10^{-5}	Residual is consistent with reference-grid discretization
Order-four stellar and local-zodiacal terms versus image sampling, 15 stars	Below 0.7% at image size 400	End-to-end check against an independent spatial discretization
Exozodiacal radial quadrature self-convergence, orders 2, 4, 6	Below 10^{-5}	Checks the one-dimensional extended-disk integration
Order-four and order-six exozodiacal terms versus image sampling	Up to about 10% on a few stars	Reference-grid aliasing near the Kennedy inner cutoff, not an order-four error: the same stars and grid deviate by 4.5–5.4% at order two
Exozodiacal radial sampling, 25-star check	Worst case 1–3% after the log-spaced substitution, against up to 35% before it	A first uniform-in-radius implementation undersampled the steep profile near the inner cutoff; the defect was found by this check and fixed
Invalid odd null order	Explicit error	Prevents applying the even-order derivation outside its domain

Table 3.2: Validation matrix for the AMS and analytic source calculation. Agreement with shared code establishes numerical consistency, not independent empirical validation

The strongest equality—AMS versus the standard source routine—is deliberately narrow: both routes share underlying physical modules. The spatial comparisons and analytic identities provide more independent numerical checks, but none validate the face-on exozodiacal assumption, the scheduler, or the fourth-order proxy as a realizable instrument.

Chapter 4

4

Trade Space Reduction

Reducing an intractable parameter space without discarding the relevant physics

The raw functional trade space cannot be explored directly. This section therefore reduces it in two steps: first by combining admissible independent random terms at one reference plane, and then by replacing spatial pixel grids with analytic integrals where source symmetry permits. Each reduction retains a stated validation boundary.

4.1 Condensation of Astrophysical Noise

The eight unrestricted modifier functions are too conditional to produce a readable engineering bridge quantity. The first reduction therefore asks which terms can be represented by one equivalent independent random-noise contribution without changing their treatment in the SNR variance.

We reduce the trade space by exploiting the additive background terms in the bin variance. We define a wavelength-dependent instrumental term $N_I(\lambda)$; in bin i, $N_{I,i} = N_I(\lambda_i)\,\Delta\lambda_i$ is added alongside the corresponding B_i term.

$$N_B(\lambda) = \sum_{i=1}^{n} N_i(\lambda) + N_I(\lambda).$$

As is evident from the blackbody behavior of the background noise sources, we must keep the wavelength dependence to maintain physical relevance. Figure 4.1 illustrates the construction: an imposed $N_I(\lambda)$ is added on top of the astrophysical background, and the shaded area between the two curves is the allowance being solved for.

The reduction is valid when terms map to a defined independent random allowance at the same reference plane; it excludes correlated/systematic terms and is not an allocation to subsystems. This study tests flat and two linear-ramp forms of N_I, not arbitrary functions. Increasing the amplitude within one specified family increases mission time; the iso-mission-time boundary then defines its tolerated amplitude.

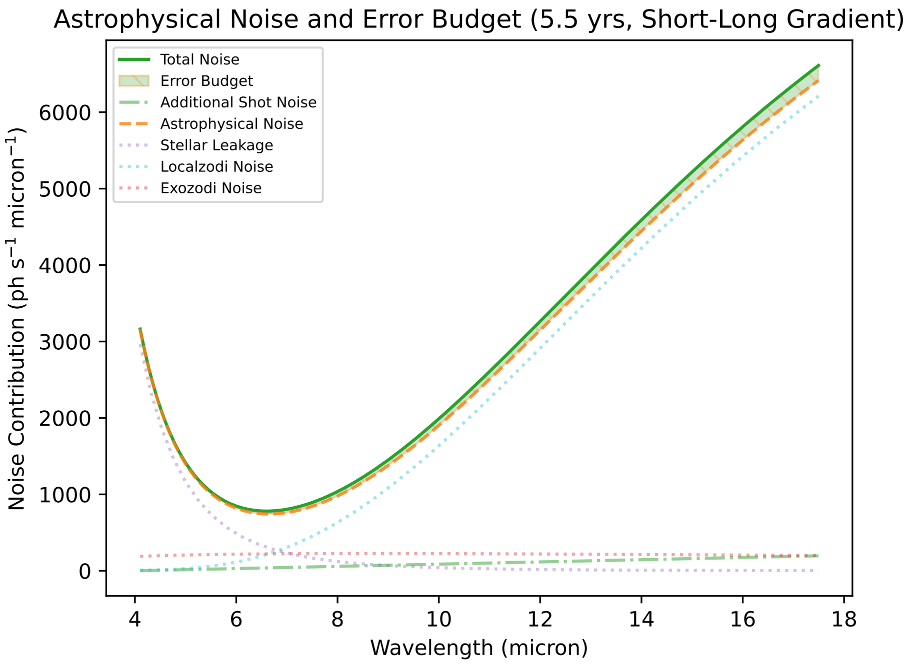


Figure 4.1: Construction of an imposed random-noise budget on top of the astrophysical background. The dashed green line is the imposed $N_I(\lambda)$, here a linear ramp that vanishes at the short-wavelength end of the band and reaches its full amplitude at the long-wavelength end; this is the long-weighted family of Section 5.2. The solid green line is the resulting total background including the budget, and the shaded area between the two curves is the allowance itself. All three curves are photon-rate spectral densities referred to the pre-efficiency single-output plane, not variances. The configuration shown is Hab2Max at null order two and the common operating point, with the amplitude set to the largest value compatible with the 5.5-year mission-time target; the same panel recurs among the results as Fig. 5.1(c).

4.2 Analytic Background Evaluation and Spectral Resolution

The analytic rewrite of the background evaluation removes the science dependence on image size. Circularly symmetric sources permit an analytic azimuthal and radial reduction of the background integrals: the stellar disk is treated as uniform and circular, local zodiacal light as locally uniform, and exozodiacal light as smooth and face-on. Inclined or asymmetric exozodiacal disks require an angle-resolved treatment and are outside this reduction. Image size now affects visualization and independent cross-checks only, not production science SNR.

The reduction follows from expanding the even power $\sin^{2p}\phi$ into cosines. Azimuthal averaging then uses the standard Bessel integral representation [15],

$$\frac{1}{2\pi}\int_0^{2\pi}\cos(z\cos\varphi)\,\mathrm{d}\varphi = J_0(z). \tag{4.1}$$

For null order $n = 2p$, azimuthally averaging one destructive output at angular radius ϑ gives

$$\langle T_{2p}\rangle_\varphi(\vartheta) = \frac{1}{2}\frac{\binom{2p}{p}}{4^p} + \sum_{j=1}^{p}(-1)^j\frac{\binom{2p}{p-j}}{4^p}J_0\left(\frac{2\pi j b\vartheta}{\lambda}\right). \tag{4.2}$$

For a uniform disk, radial area averaging replaces each $J_0(jx)$ term by $2J_1(jx)/(jx)$, following the standard Bessel derivative identity [15], where $x = 2\pi bR/\lambda$ and R is the angular disk radius. The limiting value is one as $jx \to 0$, so the expression is evaluated with its analytic limit near the origin. The Bessel power series then gives the small-star limits

$$\langle T_2\rangle_{\text{disk}} = \frac{x^2}{32} + \mathcal{O}(x^4), \tag{4.3}$$

$$\langle T_4\rangle_{\text{disk}} = \frac{x^4}{256} + \mathcal{O}(x^6). \tag{4.4}$$

Taking the ratio of Eqs. 4.3 and 4.4, the fourth-to-second-order stellar-leakage ratio therefore approaches $x^2/8$. The same general average is used inside one-dimensional radial quadrature for a tapered source or exozodiacal profile. Rotation does not "average away" a circular background; rather, circular symmetry makes its integrated response independent of rotation angle [9, 2]. Its mean may cancel after chopping, but its photons still set the shot-noise variance.

The local-zodiacal correction arose from a geometric inconsistency in the removed pixel-grid path. That path is inherited LIFEsim code and predates this work; the analytic reformulation did not introduce the defect, it exposed it. The physical field of view is circular, with solid angle proportional to πR_{FoV}^2, but one normalization divided the integrated circular flux by the area $4R_{\text{FoV}}^2$ of its enclosing square. It therefore suppressed the uniform foreground by $\pi/4$.

The decisive evidence that this is a defect rather than a modelling choice is internal to the inherited code: the untapered branch of the same routine normalized by a circular mask, while the default tapered branch normalized by the square pixel sum, so the two branches disagreed with each other on the same physical configuration. Restoring the circular normalization multiplies the local-zodiacal rate by

$$\frac{4}{\pi} = 1.2732\ldots \tag{4.5}$$

an increase of 27.3% in the limit of an untapered field of view, which is the case Appendix A derives. The production configuration applies a Gaussian field-of-view taper, and there the factor is not exactly $4/\pi$: both paths carry the same solid-angle prefactor, so the correction is the ratio of two averages of the transmission, one over a rotationally symmetric disk and one over the square evaluation grid whose corners break that symmetry. Measured against the old grid and extrapolated in resolution, that ratio converges to approximately 1.267 rather than to 1.273, and the difference is the taper rather than discretization noise. The correction is therefore an increase of about 27%, with $4/\pi$ as its untapered limit rather than its exact value. This is a normalization correction, not a new dust model.

Because local-zodiacal shot noise is present for every target, the change propagates nonlinearly through SNR, target ranking, and the 90th-percentile mission time. Comparing the corrected and uncorrected paths over every qualifying target, with the code otherwise identical, lowers the one-hour signal-to-noise ratio by a median of 9.2% at null order two and 9.8% at order four for Hab2Max, and by 9.1% and 9.8% for Hab2Min. The distribution is skewed rather than tight: the sixteenth-to-eighty-fourth percentile range runs from about −11% to −2.7%, the small shifts belonging to targets whose noise budget the local-zodiacal term does not dominate. The scope of that figure should be read carefully: it applies to results computed through the default tapered local-zodiacal path, and the present work does not attempt to propagate it into any previously published LIFE yield number. Doing so would require rerunning those studies and is left to future work.

Within this study the effect can be measured directly, and it is far larger than the median shift in signal-to-noise ratio suggests. Restoring the analytic $\pi/4$ factor reproduces the pre-correction path to within the small margin by which that factor differs from the realized ratio, so the two configurations differ in the defect and in nothing else of consequence. Table 4.1 reports both quantities the thesis is built on. Under-counting the local-zodiacal foreground shortens the mission time required with no added budget by 0.50 to 0.84 yr, between 10 and 16%. The tolerated flat allowance, however, changes by factors of 1.5 to 11.

The disproportion is the important part. A 27.3% error in a single background term produced up to an order-of-magnitude error in the derived requirement, because the amplitude is defined where the completion curve is steep: suppressing part of the background shortens the zero-budget time and leaves far more headroom before the target is reached. The Hab2Min second-order case is the extreme, where a corrected zero-budget time of

		Zero-budget time [yr]		Flat allowance [$\mathrm{ph\,s^{-1}\,\mu m^{-1}}$]		
Catalog	Null order	Pre-correction	Corrected	Pre-correction	Corrected	Ratio
Hab2Max	2	4.82	5.32	752	142	5.3
Hab2Max	4	3.86	4.47	994	641	1.6
Hab2Min	2	6.57	7.41	686	62	11.0
Hab2Min	4	5.20	6.04	897	597	1.5

Table 4.1: Effect of the inherited local-zodiacal normalization defect on the two quantities this study reports, at the common operating point and each catalog's primary mission-time target. The pre-correction column is obtained by restoring the suppression factor, so the two configurations differ only in the defect. Amplitudes are the values returned by the search rather than the rounded entries of Table 5.1; Appendix B.3 records the difference between the two.

7.41 yr against a 7.5 yr target leaves almost no room, and the allowance collapses from 690 to 60. Had the defect gone unnoticed, the requirements reported here would have been up to an order of magnitude too permissive.

Transmission and source spectra are integrated within every wavelength bin using an eight-point Gauss–Legendre rule in inverse wavelength $u = 1/\lambda$. This coordinate is natural because interferometric phases are proportional to b/λ and therefore vary nearly linearly with u. Across spectral resolutions 5, 10, 20, and 50 and sampled catalog stars, the stellar-leakage and exozodiacal terms agree with a 64-point reference to approximately machine precision. The adopted resolving power remains $R = \lambda/\Delta\lambda = 20$, which produces 31 bins over 4–18.5 μm. That choice is a compute-accuracy trade rather than a physical one, and Figure 4.2 shows both sides of it. The compute cost of a full catalog pass grows steadily with R, since the number of bins sets the width of every per-target source evaluation. The accuracy side is regime-dependent. Referred to an $R = 100$ evaluation, the SNR computed at $R = 20$ is converged to a few parts in 10^4 when the astrophysical background sets the bin variance, but departs by orders of magnitude more once the modeled per-detector and thermal instrument-noise terms dominate it. The production configuration used here is the former case: those instrument-noise modules are deliberately absent, since representing them by an aggregate allowance is precisely the reduction of Section 4.1, so the reported results sit in the converged regime. The figure does not establish how an imposed N_I of comparable magnitude would behave, because its spectral shape is prescribed rather than thermal; that remains a boundary of the retained binning rather than a demonstrated error.

Thus spatial image size is no longer a science parameter, while wavelength-bin integration is explicitly converged for the retained binning. Appendix A records the derivation and numerical conventions in greater detail.

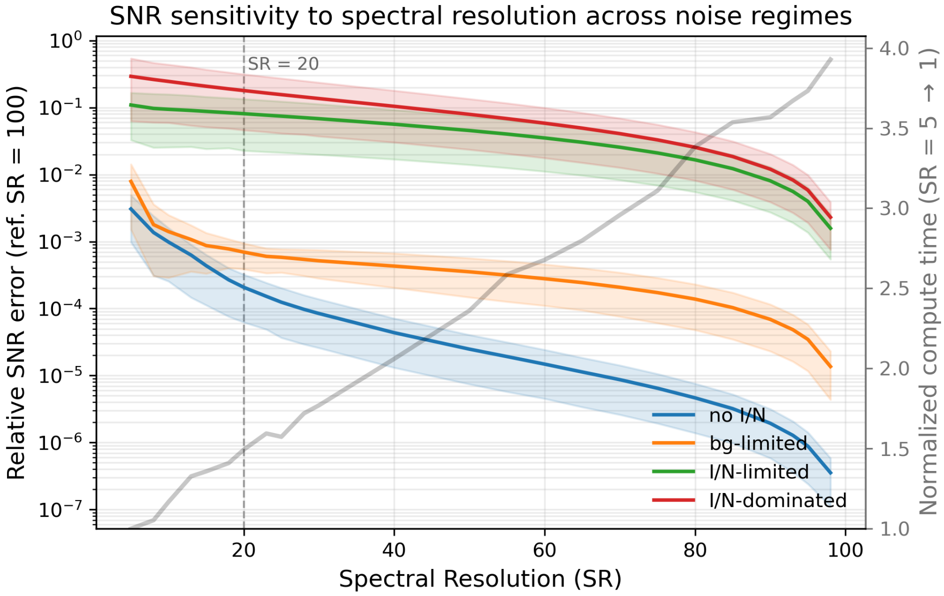


Figure 4.2: Sensitivity of the computed SNR to spectral resolving power, referred to an $R = 100$ evaluation, for four noise regimes: no instrumental term, background-limited, instrument-limited, and instrument-dominated. Solid lines are medians over sampled catalog targets; shaded bands show their spread. The grey curve, read against the right axis, is the compute time of a full pass normalized to $R = 5$. The dashed line marks the adopted $R = 20$.

4.3 Restriction of Architecture

The transmission model of Section 3.3 accepts any combiner, which makes the space of architectures as unbounded as the space of noise functions was before the first reduction. This study restricts it to two: the reference double Bracewell, and the six-aperture double triple nuller of Section 3.4 as a realizable fourth-order design. The Angel Cross is excluded not by choice but by construction, since it offers a single deep output and therefore cannot chop.

The restriction is deliberate and it bounds the conclusions. Each architecture is evaluated at its own optimal baseline rather than at the reference's, so the comparison is between designs each doing the best it can, but two designs are not a survey. Total collecting area is held fixed between them, so the six-aperture design is six smaller collectors rather than six of the same size; whether that is the right systems-level comparison is a mission-design question this thesis does not settle.

Chapter 5

5

Trade Space Exploration

From the reduced parameter space to an iso-mission-time random-noise allowance

With the trade space reduced, we now invert the AMS mapping. All results use the analytic background model of Section 4.2 and a deliberately conservative common operating point unless a plotted axis varies it. The working LIFE field-of-regard reference is 90°; 65° is substantially more restrictive but still permits acceptable mission times. Current project estimates place the K-band limiting magnitude near 8; magnitude 7 is a pessimistic lower bound, and the completed scans show little remaining sensitivity near that value. A 10 h slew is the inherited baseline [7]; 12 h adds an intentionally simple 20% margin, while longer slews rapidly remove feasible budget space. The choices are therefore conservative comparison points, not optimized requirements. Hab2Max is evaluated at study targets of 5.5 and 6 yr; Hab2Min uses 7.5 and 8 yr. These targets are chosen, not inherited. The LIFE concept is postulated to run for roughly five years, so a shorter target is the more demanding and therefore the more informative requirement; for each catalog the lower target is the smallest half-year step at which a non-trivial positive allowance still exists at the nominal null order. Below it the zero-budget calculation already exceeds the requested time and no allowance can be reported. The second target adds half a year to expose how steeply the allowance responds to a relaxation of the science schedule. Because the two catalogs reach that feasibility threshold at different times, they are not compared at a common mission time; differences between Hab2Max and Hab2Min in Table 5.1 therefore combine the occurrence scenario with the different target, and should not be read as an occurrence effect alone. Appendix B records this operating point together with the data provenance, the reproduction of the reported amplitudes, and the statistical interpretation of the sampled percentile.

5.1 The Iso-Mission-Time Budget

The *iso-mission-time budget* is the tolerated scalar amplitude of one imposed $N_I(\lambda)$ shape at a stated operating point. It is expressed in $\mathrm{ph\,s^{-1}\,\mu m^{-1}}$ for the flat case or as a ramp endpoint. "Below" applies only to the same imposed shape and reference calculation.

The search starts with an amplitude bracket from 0 to 50,000 $\mathrm{ph\,s^{-1}\,\mu m^{-1}}$. At every iteration the lower bracket endpoint is admissible and the upper endpoint exceeds the target mission time. An exponential fit through the endpoint evaluations proposes the next amplitude; it accelerates the search but does not define the answer. The new evaluation replaces one endpoint according to its mission time, so the bracket invariant is retained. The search stops when mission time lies within 0.05 yr of the target or the amplitude bracket is at most $10\,\mathrm{ph\,s^{-1}\,\mu m^{-1}}$ wide. Reported amplitudes are therefore search point estimates; Section B.3 quantifies the resulting uncertainty by rerunning the complete search and recording the mission time each reported amplitude actually delivers. The two alternative stopping rules do not support a universal ± 5 amplitude uncertainty: a time-tolerance stop may leave a wider amplitude bracket, whereas a bracket-width stop only bounds the final numerical search interval.

Within a fixed target list, added independent variance can only lower individual SNRs. The full mission-time map also includes discrete rankings and a sampled percentile, so global monotonicity is an empirical property rather than a formal theorem. The completed scans showed nested contours and no folded iso-time branches at the sampled resolution. If the zero-budget calculation already exceeds the requested time, no non-negative allowance exists; the plotted dark region represents this physically infeasible operating point, not a failed numerical search.

5.2 The Shape of the Error Budget

The first question is how the affordable budget depends on its *distribution* over wavelength. We compare three one-parameter families:

$$N_{\text{flat}}(\lambda) = A, \tag{5.1}$$

$$N_{\text{short}}(\lambda) = A\,\frac{18.5\,\mu\text{m} - \lambda}{14.5\,\mu\text{m}}, \tag{5.2}$$

$$N_{\text{long}}(\lambda) = A\,\frac{\lambda - 4\,\mu\text{m}}{14.5\,\mu\text{m}}. \tag{5.3}$$

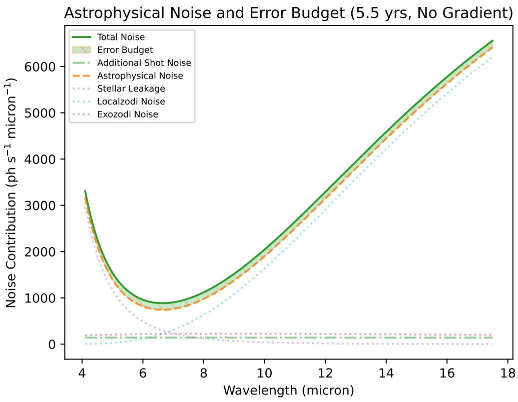
Here A is the reported amplitude. The ramp endpoints are imposed shape parameters, not independently inferred wavelength limits. Figures 5.1 and 5.2 show the Hab2Max 5.5-year solutions.

Catalog	Mission-time target [yr]	Null order	Flat	Short-weighted	Long-weighted
				Amplitude [ph s^{-1} μm^{-1}]	
Hab2Max	5.5	2	140	300	210
Hab2Max	5.5	4	640	1080	1470
Hab2Max	6.0	2	610	1300	1090
Hab2Max	6.0	4	940	1830	2240
Hab2Min	7.5	2	65	200	150
Hab2Min	7.5	4	590	1000	1260
Hab2Min	8.0	2	460	970	810
Hab2Min	8.0	4	790	1460	1860

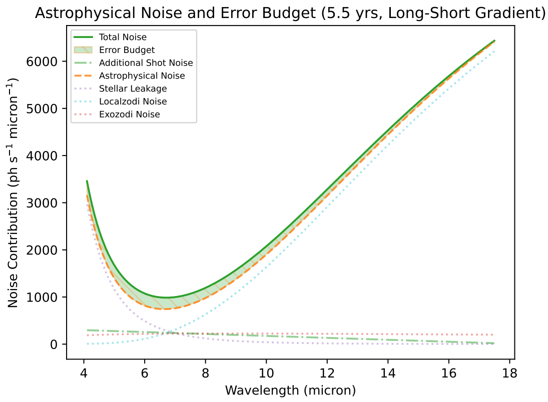
Table 5.1: Final amplitudes at the common operating point. A ramp value is its non-zero imposed endpoint. These endpoint parameters compare sensitivity within the tested families; they are not common integrated-noise measures or engineering-cost rankings.

The ordering is consistent across catalogs and both Hab2Max mission-time targets. At null order two, the short-weighted endpoint is largest. At null order four, the long-weighted endpoint is largest. Relaxing the target time also increases the allowance sharply: for example, the Hab2Max flat budget grows from 140 to 610 at order two and from 640 to 940 at order four when the target changes from 5.5 to 6 yr. For Hab2Min, the corresponding flat allowances rise from 65 to 460 and from 590 to 790 when the target changes from 7.5 to 8 yr.

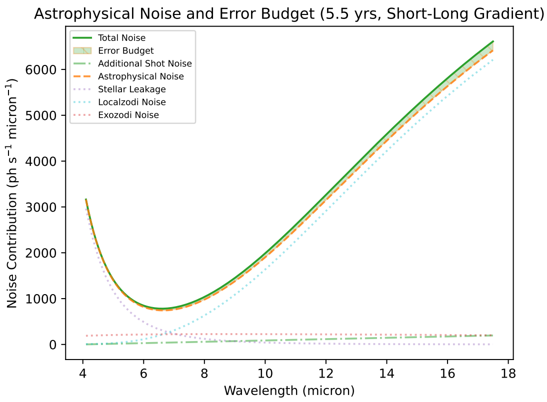
A ramp endpoint and a flat amplitude are not directly comparable quantities, so it is worth checking that the reversal is not an artifact of the chosen reporting convention. A ramp rising linearly from zero to A across the band has mean $A/2$, so a flat budget of $A/2$ adds the same band-integrated noise; the integral-neutral ramp endpoint is therefore twice the flat amplitude in the same row. Measured against that reference, the short-



(a) Flat



(b) Short-weighted



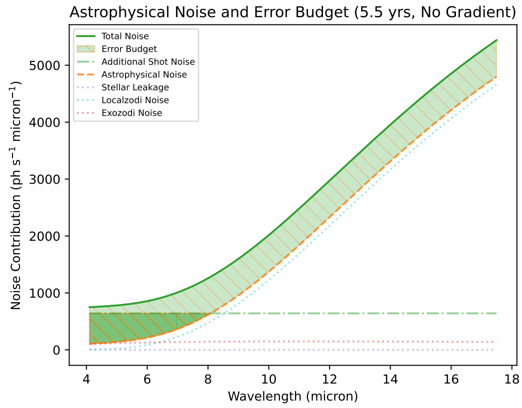
(c) Long-weighted

Figure 5.1: Final null-order-two Hab2Max budgets at the common operating point and 5.5-year target. Stellar leakage dominates the short-wave astrophysical background, so the short-weighted ramp admits the largest endpoint. The titles inside the panels name each ramp by the direction it travels across the band, whereas the subcaptions name it by the end that carries the allowance; a ramp labelled "Long-Short" inside a panel rises towards short wavelengths and is therefore the short-weighted family.

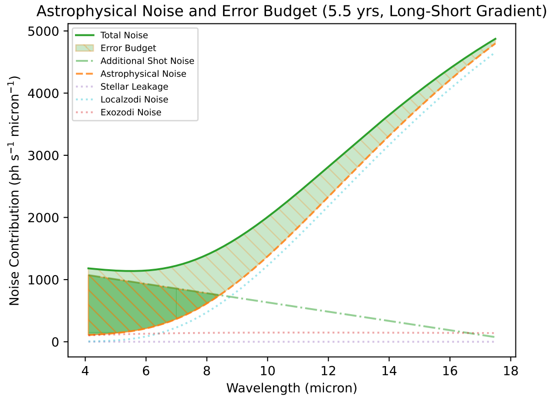
weighted endpoint exceeds the neutral value at null order two in all four rows, and the long-weighted endpoint exceeds it at null order four in all four rows. The reversal therefore holds under equal integrated added noise, not only under equal endpoints.

The strong response to a half-year relaxation is not evidence of linear sensitivity. These target times lie near a steep part of the completion curve, where a small SNR loss can move marginal planets below the selected set and where the 90th-percentile universe can change identity. The budget endpoints should therefore be interpolated only through a fresh AMS inversion, not by scaling the values in Table 5.1.

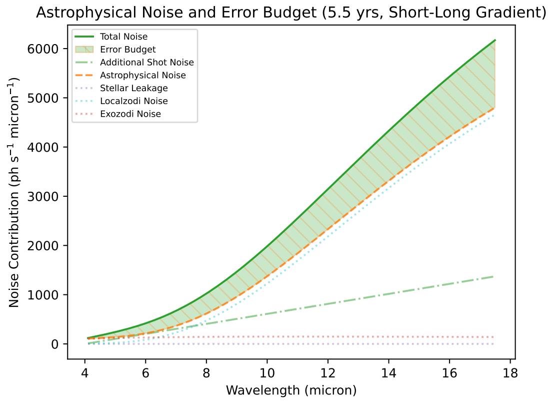
The two-dimensional scans in Fig. 5.3 confirm that the inversion is not an artifact of comparing only the axis endpoints. At order two, equal-mission-time contours extend farther along the short-wavelength-budget axis. At order four, they extend farther along



(a) Flat



(b) Short-weighted



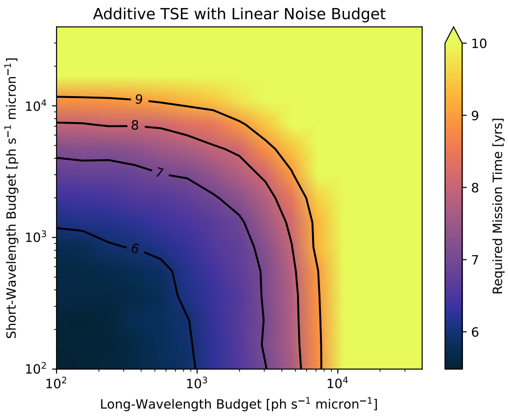
(c) Long-weighted

Figure 5.2: Final null-order-four proxy budgets for the same Hab2Max operating point and target as Fig. 5.1. Geometric stellar leakage is nearly absent. The long-weighted ramp now admits the largest endpoint because the remaining background is dominated by zodiacal emission toward long wavelengths. Panel titles follow the same direction-of-travel convention noted for Fig. 5.1.

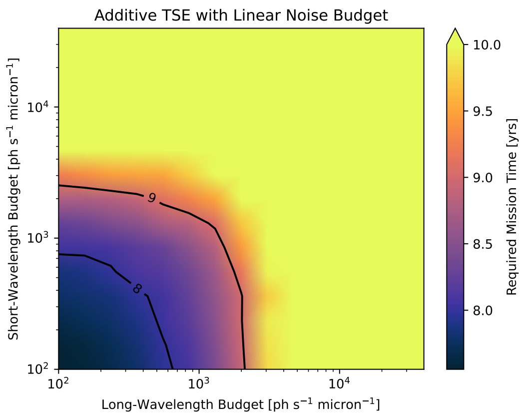
the long-wavelength-budget axis.

5.3 The Random-Noise Error Budget

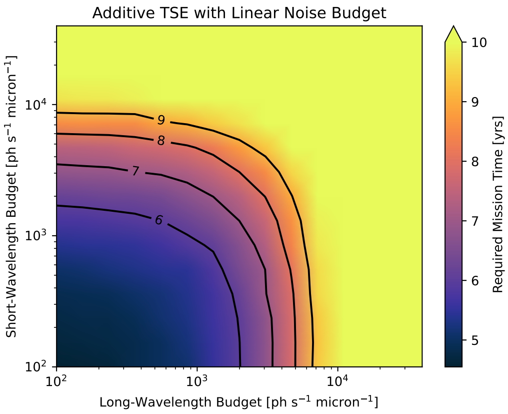
Table 5.1 is the numerical answer to the central research question for the tested families and operating points. For any chosen row and shape, the reported value is the amplitude at which the search terminated, delivering a mission time within 0.05 yr of the target. That achieved time may lie slightly above or slightly below the target, as Table B.2 records, so a reported amplitude should be read as the boundary located to within the search tolerance rather than as a guaranteed admissible bound. The flat family is the most directly comparable scalar allowance. The ramps test where added random variance is least damaging, but their unequal spectral integrals mean that their endpoint amplitudes



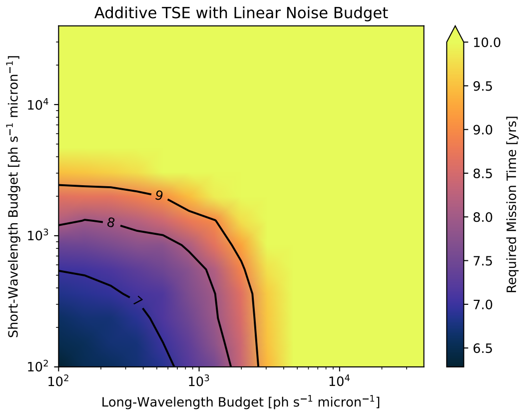
(a) Order 2, Hab2Max



(b) Order 2, Hab2Min



(c) Order 4, Hab2Max



(d) Order 4, Hab2Min

Figure 5.3: Required mission time over independent short- and long-wavelength ramp endpoints at magnitude 7, FoR 65°, and slew 12 h. Black curves mark integer mission-time contours. The preferred axis reverses between null orders in both catalogs.

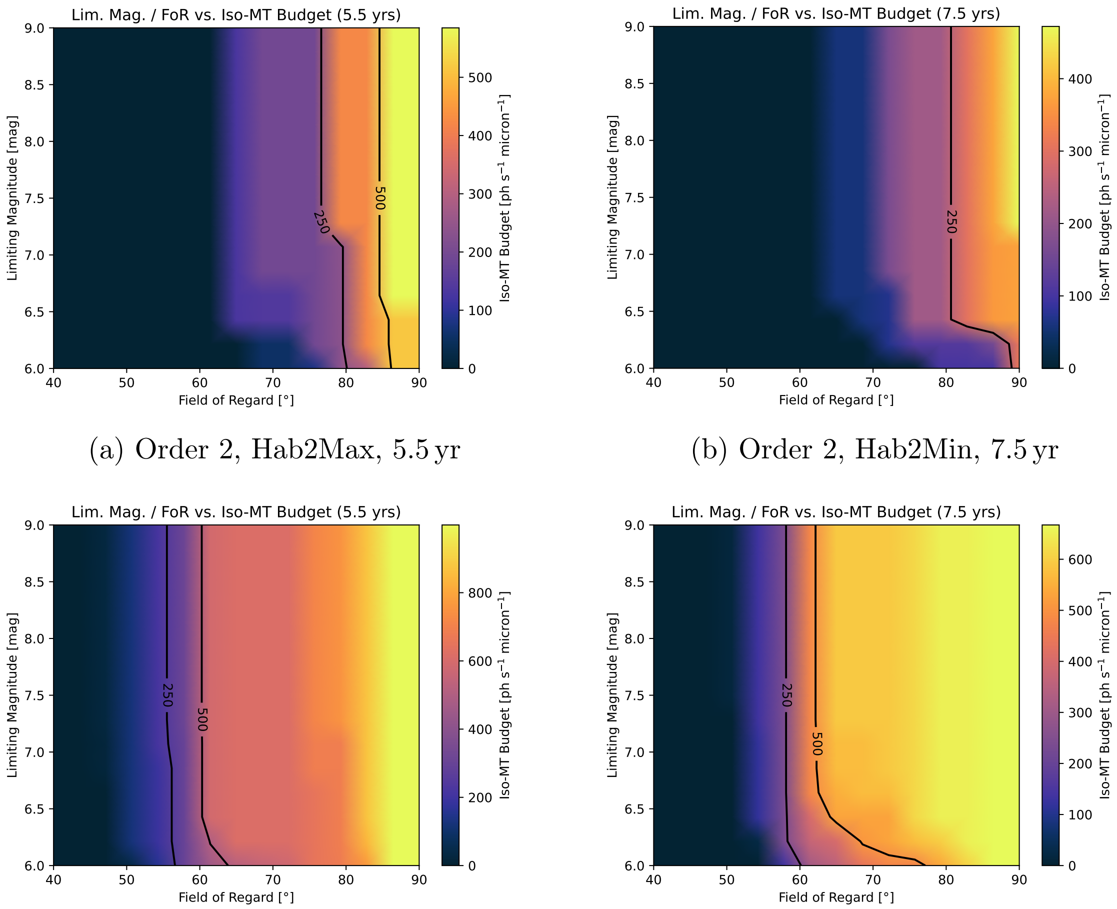
must not be interpreted as a global cost ranking.

The result also does not provide independent limits at individual wavelengths. For example, the zero at one end of a ramp is imposed by its definition, not inferred from the experiment. Converting these aggregate allowances into an instrument requirement requires a separate allocation model that maps subsystem noise terms to the same reference plane and combines their independent variances.

5.4 Dependence on the Agnostic Parameters

The remaining scans use the flat budget and vary two agnostic parameters at a time. They therefore show operating-point dependence of the flat allowance and cannot automatically be transferred to either ramp family.

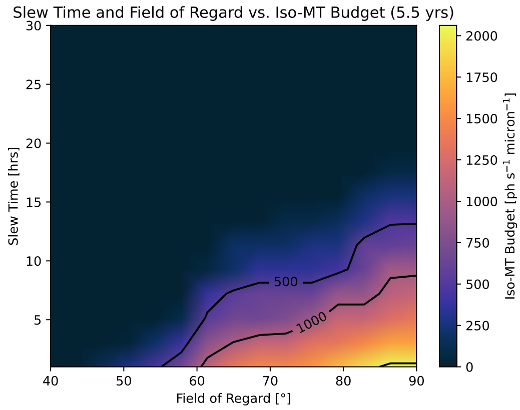
Figure 5.4 varies limiting magnitude and field of regard at fixed 12 h slew time. In all four panels, widening the field of regard has the larger effect because it admits more potential targets. The magnitude dependence largely saturates near the adopted value of 7, especially for Hab2Max: once most high-SNR qualifying stars pass the magnitude cut, admitting still fainter targets adds little to the optimized top-50 sample. Field of regard acts differently because it removes entire sky regions irrespective of target brightness. The fourth-order proxy generally permits a larger flat allowance, but the same qualitative filter dependence remains.



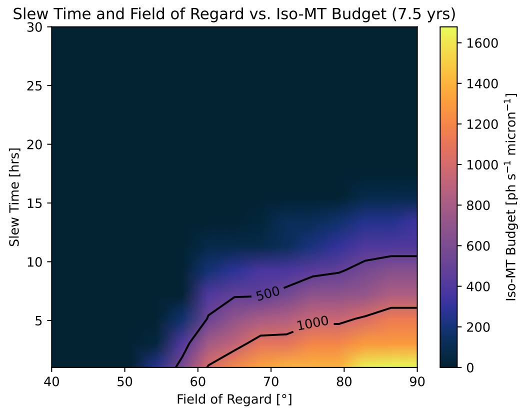
(a) Order 2, Hab2Max, 5.5 yr
(b) Order 2, Hab2Min, 7.5 yr
(c) Order 4, Hab2Max, 5.5 yr
(d) Order 4, Hab2Min, 7.5 yr

Figure 5.4: Final iso-mission-time flat-budget scans over limiting magnitude and field of regard, with slew time fixed at 12 h. Dark regions have no admissible positive budget at the corresponding operating point.

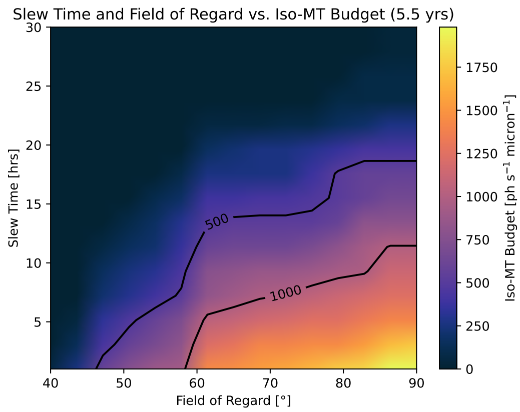
Figure 5.5 varies slew time and field of regard at fixed limiting magnitude 7. Longer slews consume time without adding SNR and therefore reduce the admissible budget. The dependence is especially important because each selected planet receives multiple follow-up visits: slew overhead is charged during the shared detection campaign, four additional orbit epochs, and characterization. Wide field of regard and short slew time therefore act together: more targets are available and less overhead is charged for moving between them.



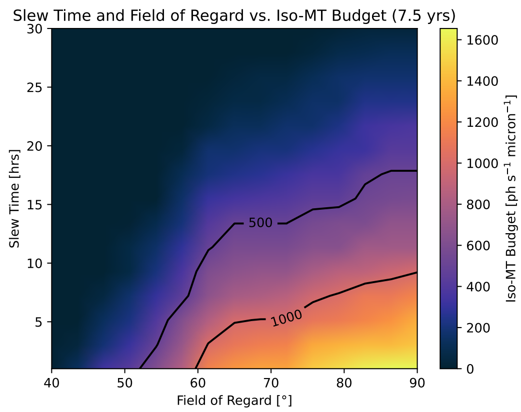
(a) Order 2, Hab2Max, 5.5 yr



(b) Order 2, Hab2Min, 7.5 yr



(c) Order 4, Hab2Max, 5.5 yr



(d) Order 4, Hab2Min, 7.5 yr

Figure 5.5: Final iso-mission-time flat-budget scans over slew time and field of regard at limiting magnitude 7. Dark regions have no admissible positive budget.

5.5 Catalog Dependence

The Hab2Min and Hab2Max calculations show the same qualitative shape ordering and the same null-order inversion. Hab2Min requires longer mission-time targets and generally returns smaller allowances at otherwise matched operating parameters, consistent with its less favorable planet population. Because the adopted mission-time targets differ between catalogs, the numerical difference cannot be attributed to $\eta_\oplus$ alone. Figure 5.6 illustrates the field-of-regard filter for one Hab2Max selection and is not evidence of catalog independence.

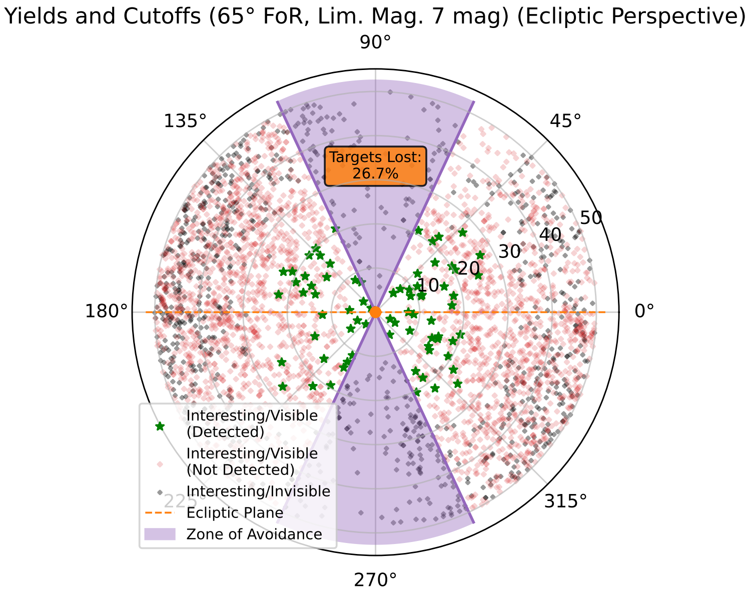


Figure 5.6: Ecliptic view of the final Hab2Max selection at magnitude 7 and FoR 65°. Here "interesting" denotes Experiment-1-eligible targets. The 26.7% value is specific to this catalog and selection.

5.6 Why the Spectral Preference Reverses at Null Order Four

Suppression of stellar leakage is the leading physical interpretation of the completed order-four comparison. Equations 4.3 and 4.4 together predict a fourth-to-second-order ratio of approximately $x^2/8$. Across the 4505 unique catalog stars, $x = 2\pi b R_\star/\lambda$ remains above 0.005 over the modeled band, outside the cancellation-sensitive numerical regime of the direct Bessel sum. At 10 µm the median leakage ratio is about 6.3×10^{-5}; at 18.5 µm it is about 1.9×10^{-5}. Direct positive quadrature agrees with the analytic order-four response over the complete catalog range.

This suppression removes the high short-wave stellar-noise floor visible in Fig. 5.1. Added short-wave variance then becomes comparatively expensive. The remaining local- and exo-zodiacal backgrounds dominate toward longer wavelengths, so a larger long-weighted endpoint can be hidden beneath their photon noise. This explanation is consistent with the reversal in Table 5.1 and Fig. 5.3; however, no ablation run isolates leakage suppression from the simultaneous extended-source and planet-throughput changes.

Table 5.2 collects the effects that change between the two orders alongside their consequences for the reported allowance. The result is physical within the proxy but not a standalone architecture prediction. Without throughput normalization, the far-field average of the null envelope changes from 1/4 at order two to 3/16 at order four. Local-zodiacal transmission therefore falls by approximately 25%, and the off-axis planet response also changes. The order-four result consequently combines deeper central suppression with a different throughput distribution while holding all other instrument parameters fixed.

Effect	Order-two reference	Order-four proxy consequence
Near-axis stellar leakage	Leading term $x^2/32$	Leading term $x^4/256$; median ratio 6.3×10^{-5} at 10 µm
Extended-source envelope average	1/4	3/16, a 25% reduction before other weighting
Dominant short-wave background	Stellar leakage	Leakage largely removed; added short-wave variance becomes more costly
Least damaging tested ramp	Short-weighted	Long-weighted
Interpretation	Double-Bracewell reference	Controlled response proxy, not a realizable architecture

Table 5.2: Synthesis of the physical and numerical changes behind the null-order inversion.

5.7 Comparing Realizable Architectures

The preceding results describe the reference array. Because the transmission model of Section 3.3 accepts any combiner, the same inversion can be run for a genuinely realizable fourth-order design and the two compared directly, each sized to its own response peak.

That last point is not a detail. The baseline prescription of Section 3.6 places the first peak of the chopped response on the habitable-zone centre, and the constant that does so is a property of the response, not of the mission: it is 0.5894 for the double Bracewell and 1.1376 for the double triple nuller. Sizing the six-aperture design with the reference constant leaves it operating at 31 % of its achievable signal. A comparison made that way measures the mismatch, not the architecture.

Table 5.3 reports the outcome at the common operating point. The fourth-order design completes Experiment 1 in less time than the reference with no added budget, by 6.4 % for Hab2Max and 6.0 % for Hab2Min, while tolerating a comparable flat allowance at the primary target.

The gain does not come from the population as a whole. Taken over every qualifying target the fourth-order design is the weaker instrument: its median one-hour signal-to-noise ratio is 0.93 of the reference for Hab2Max and 0.90 for Hab2Min, and only about 28 % of targets improve at all. The improvement is concentrated where it matters. Ranking targets by their reference signal-to-noise ratio, the best fifty—the number Experiment 1 observes—improve by a median factor of 1.22 and 1.20 respectively, the best five hundred

Catalog	Architecture	Zero-budget time [yr]	Flat	Long-weighted
Hab2Max	Double Bracewell	5.32	142	210
Hab2Max	Double triple nuller	**4.98**	138	**347**
Hab2Min	Double Bracewell	7.41	62	157
Hab2Min	Double triple nuller	**6.96**	129	**308**

Table 5.3: Reference and fourth-order architectures at the common operating point, each sized to its own response peak. Amplitudes are in $\mathrm{ph\,s^{-1}\,\mu m^{-1}}$ at each catalog's primary mission-time target, 5.5 yr for Hab2Max and 7.5 yr for Hab2Min. Total collecting area is equal between the two designs.

by about 1.15, and the lower half of the distribution degrades to 0.85 and 0.80.

The mechanism is the one the null order was meant to exploit. The best targets are the nearest and brightest stars, which are also the most resolved, so they suffer most from the geometric stellar leakage that a second-order null cannot suppress. The deeper null removes it. Faint distant targets are background-limited instead, where a lower far-field transmission costs and the absent leakage was never the constraint. A mission observes the top of the distribution, not its median, and that is why a design that is worse on average completes the programme sooner.

The spectral preference established in Section 5.6 survives the substitution. Across both catalogs and both mission-time targets the long-weighted family remains the most tolerant, by factors of 1.28 to 1.91, and exceeds the integral-neutral reference of twice the flat amplitude in every case. The reversal is therefore a property of a buildable architecture and not of an idealized response.

Two boundaries limit the comparison. The equal-area convention means the six-aperture design is six smaller collectors rather than six of the reference size, and whether that is the appropriate systems-level trade is not settled here. And two architectures are a comparison, not a survey: Eq. 3.4 admits an arbitrary combiner, and the space of designs that satisfy the chopping requirement has not been explored.

Chapter 6

6

Discussion

Interpreting the allowances, the null-order inversion, and their boundaries

The thesis asks two connected questions: how much aggregate random noise LIFE can tolerate at a chosen mission time, and how that allowance changes when the modeled central null becomes steeper. The completed results answer both within the controlled reference calculation before the methodological and architectural boundaries are considered.

6.1 The Error Budget and Architecture Comparison

At the adopted operating point, the nominal second-order null permits flat allowances of $140\,\mathrm{ph\,s^{-1}\,\mu m^{-1}}$ for Hab2Max at 5.5 yr and 65 for Hab2Min at 7.5 yr. Within the tested ramp families, short-weighted noise is least damaging because it is added where stellar leakage already dominates. This finding is specific to the second-order background hierarchy.

An allowance in photons per second per micron carries no sense of scale on its own, so it is worth stating what it represents against the background it is added to. Taking the median over catalog stars at the same reference plane, the astrophysical background averages about $4300\,\mathrm{ph\,s^{-1}\,\mu m^{-1}}$ across the band at null order two and about 2350 at order four. Two distinct effects produce that reduction. The deeper null lowers the far-field transmission seen by every background term, exactly and measurably: the uniform local-zodiacal contribution scales between the orders by 0.7500, which is precisely the ratio 3/16 to 1/4 of the far-field averages quoted in Section 5.6. Separately, and only for the resolved stellar disk, the x^4 dependence removes the leakage almost entirely.

Attributing the whole reduction to suppressed stellar leakage would therefore be wrong, although leakage is what dominates it at the short-wavelength end where x is largest. The second-order flat allowances therefore amount to roughly 3% of the natural background for Hab2Max and 1.5% for Hab2Min: the instrument must be quiet to within a few percent of an irreducible foreground, which is a demanding requirement. At order four the same allowances are about a quarter of the reduced background, which is a far more comfortable one.

The composition of that background also explains why the normalization correction of Section 4.2 propagated so strongly. Near $10\,\mu\mathrm{m}$ the local-zodiacal term supplies roughly 1760 of the $2260\,\mathrm{ph\,s^{-1}\,\mu m^{-1}}$ present at order two, and about 1320 of 1530 at order four, so it is the dominant contribution in both cases. A 27.3% error in the largest single term of a background against which a few-percent requirement is being set is not a small perturbation. Over the same interval the stellar-leakage term falls from about 184 to $0.011\,\mathrm{ph\,s^{-1}\,\mu m^{-1}}$ between the two orders, a suppression of roughly four orders of magnitude, consistent with the $x^2/8$ scaling of Section 5.6.

A deeper null changes that hierarchy. For the six-aperture design of Section 5.7 the long-weighted ramp becomes the least damaging family, in both catalogs and at every mission-time target. The interpretation is geometric: the fourth-order response suppresses geometric stellar leakage by roughly four orders of magnitude and removes the short-wave variance floor, so added noise at short wavelengths is no longer hidden beneath a background that was there anyway, while the rising zodiacal terms at long wavelengths still hide it.

What the architecture comparison adds is that this reordering survives the move from an idealized response to a buildable one, and that the accompanying change in mission time runs opposite to intuition. The fourth-order design is the weaker instrument for most of the catalog, with a median one-hour signal-to-noise ratio of about 0.9 of the reference, yet it completes Experiment 1 sooner. The resolution is that a mission observes the top of the target distribution rather than its median: the best fifty targets, which are the nearest and most resolved stars and therefore the ones most limited by geometric leakage, improve by about a fifth. A design that is worse on average is better where the schedule is actually set.

This is also where an idealized comparison would have misled. Holding throughput, collecting area and the baseline prescription fixed while deepening the null describes no realizable instrument, and doing so overstates the fourth-order allowances by a factor of several and places the feasibility boundary in field of regard some fifteen degrees too far. The quantity that survives is the ordering of the budget families; the magnitudes do not.

6.2 The Agnostic Approach and the AMS

The methodological contribution is mechanism-agnostic treatment of an *aggregate* independent random-noise allowance, rather than a subsystem allocation. This common reference-plane quantity allows science completion time to constrain a later engineering allocation without prematurely choosing the physical origin of every random term.

The AMS separates its SNR and filter flows (Section 3.6). Magnitude, field-of-regard, and slew changes therefore reuse cached unfiltered SNRs, while physical or budget changes trigger recomputation. Host-star memoization further avoids repeated source calculations for planets sharing one star. These boundaries turn the repeated evaluations required by the TSE from an impractical uniform grid into targeted inversions.

6.3 What the Reductions Buy

The first reduction combines only independent random terms that can be referred to the same single-output plane as $N_I(\lambda)$ (Section 4.1). The second replaces spatial-grid evaluation of symmetric backgrounds with exact azimuthal averages and one-dimensional radial quadrature. Together they reduce a formally infinite-dimensional problem to explicit scalar searches, remove image size as a production-science parameter, expose the local-zodiacal normalization error, and provide the analytic null-order scaling used to interpret the result.

That last exposure carries a caution that outlives this study. An iso-mission-time requirement is defined where the completion curve is steep, so it inherits an amplified version of any error in the background it is measured against: a 27.3% error in one term moved the derived allowance by up to a factor of eleven (Table 4.1), and the amplification is largest exactly where the target is most demanding and the result therefore most interesting. Requirements derived this way should be quoted with the background model that produced them, and a background correction should be treated as invalidating them rather than perturbing them.

6.4 Limitations

Table 6.1 ranks the remaining boundaries by their ability to change the numerical allowance or its engineering interpretation.

The first limitation sets the strongest boundary on the central comparison. Order four is an impact study of a steeper central null under retained assumptions, not a design pre-

Priority	Limitation	Consequence
1	Unnormalized null-order proxy	The order-four comparison combines deeper suppression with changed planet and extended-source throughput; it is not a design prediction.
2	Scheduler boundary behavior	Strict completion tests require at least 51 detections in at least 91 universes, while empty universes are omitted. These choices can bias mission time in opposite directions.
3	Aggregate random-noise model	Correlated instability, calibration bias, and systematic error cannot be allocated from N_I and may impose stricter requirements.
4	Sampling uncertainty	The 100 universes resolve only a sampled 90th percentile. Its 95.6% distribution-free rank interval spans order statistics 84–96, but saved outputs do not retain the corresponding time amplitudes.
5	Symmetric exozodiacal model	Inclined, clumpy, or asymmetric disks can create rotation-dependent structure absent from the face-on radial reduction.
6	Tested spectral families	Flat and two linear ramps sample three directions in a functional trade space; endpoints are not independent wavelength limits or engineering-cost rankings.

Table 6.1: Ranked limitations of the reported error budgets. Priority reflects relevance to the thesis's central numerical and architectural claims, not ease of remediation.

diction. It provides no realizable beam-combination matrix, aperture count, optical-loss model, instability-noise calculation, or control requirement. Likewise, the point estimates inherit both search tolerance and finite-universe rank uncertainty; neither is recoverable as a single symmetric error bar from the retained products.

6.5 Outlook

The next scientific step is not another run of the present trade space, but a more physical mapping. A realizable higher-order combiner should replace the $\sin^n$ proxy, including its aperture geometry, normalized throughput, optical losses, and instability terms. The aggregate allowance should then be allocated to concrete subsystems at the same reference plane.

Before another production campaign, the scheduler should use non-strict completion criteria consistent with the stated experiment, retain failed universes explicitly, and save every per-universe component time. This would permit bootstrap or rank-based uncertainty intervals in years and separate detection, orbit, characterization, and slew sensitivities. Extending the analytic framework to inclined or asymmetric exozodiacal disks would remove the next important symmetry assumption. These steps would turn the present defensible impact study into a stronger architecture-level requirement analysis.

Chapter 7

7

Conclusion

What the completed simulations establish—and what remains architecture-dependent

The completed AMS and TSE answer the central research question for every tested budget family and operating point. Under the analytic background model of Section 4.2, the nominal second-order null permits primary flat allowances of 140 $\mathrm{ph\,s^{-1}\,\mu m^{-1}}$ for Hab2Max at 5.5 yr and 65 for Hab2Min at 7.5 yr. The order-four proxy raises them to 640 and 590, while the least damaging tested ramp reverses from short-weighted to long-weighted.

The sub-questions close as follows:

- The unrestricted AMS trade space contains up to eight free modifier functions and three scalar agnostic parameters and is therefore formally infinite-dimensional. The reported family searches reduce it to one amplitude dimension, four dimensions when all agnostic parameters vary, or five when two ramp endpoints vary independently.
- Magnitude, field-of-regard, and slew changes reuse unfiltered SNRs and repeat only filtering and scheduling. Budget, source-physics, or null-order changes require catalog-wide SNR recomputation, although host-shared source terms remain memoized within that pass.
- The two reductions must be judged separately, and only one of them is validated here. The *numerical* reduction, analytic integration of the circularly symmetric backgrounds in place of a spatial pixel grid, removes the grid dimension and reproduces the retained source calculation to the precision stated in Table 3.2. The

modelling reduction, representing all admissible independent random terms by a single aggregate $N_I(\lambda)$ at a common reference plane, removes the unbounded functional dimension and is exact for terms that are genuinely independent and referable to that plane, because independent variances add. What is not tested is how large a fraction of a realistic instrumental noise inventory satisfies that condition. Quantifying the departure for a specific decomposition into subsystem terms is left open, and the sub-question is therefore answered for the numerical reduction and only partially for the modelling one.

- The TSE locates maximum flat and linear-ramp amplitudes as iso-mission-time point estimates; Table 5.1 reports the complete set, including both catalogs and both mission-time targets.
- The response of an arbitrary nulling architecture can be evaluated at the cost of the closed-form reference. Writing the transmission in terms of aperture positions and a combiner matrix, and exploiting the fact that the rotation average of any single-baseline geometry is a one-dimensional function of $\pi b\theta/\lambda$, gives a full evaluation in 14.3 s against 14.7 s for the hand-derived response, agreeing with it to a relative 6×10^{-12}. The simulator is therefore architecture-agnostic in the sense the central question requires, and not at the price of the efficiency the same question demands.
- A fourth-order central null suppresses geometric stellar leakage by about four orders of magnitude and reverses the spectral preference toward long wavelengths, in both catalogs, at every mission-time target, and for a realizable six-aperture combiner as well as for an idealized response. The ordering is thus a property of the null depth rather than of the modelling convention.
- The requirement does depend on the architecture, and not only in magnitude. At equal collecting area the six-aperture fourth-order design completes Experiment 1 about six percent sooner than the reference while tolerating a comparable flat allowance, and it reaches operating points in field of regard that the reference cannot. It is nevertheless the weaker instrument for most of the catalog; the gain is concentrated in the best fifty targets, which is the number the experiment observes. A deeper null therefore buys reach rather than margin (Section 5.7).
- A constraint follows from the combiner structure alone rather than from any simulation. Chopping requires two deep outputs of opposite modulation sign, and a four-aperture array admits only one fourth-order output. A fourth-order null is therefore incompatible with LIFE's chopping scheme at four apertures, and six are the minimum at which both are available.

These allowances do not yet constitute subsystem requirements. They define a validated bridge from scientific yield to an explicit engineering trade space and show which spectral regions most strongly control mission time. That bridge matters because LIFE's promise—new knowledge of nearby terrestrial planets and the possibility of life beyond Earth—depends not only on asking an ambitious question, but on demonstrating that an instrument can answer it within credible limits.

Appendix A

A

Analytic Transmission Derivation

Derivation of the circularly symmetric transmission and background response

This appendix records the reduction used by the production source model. It is included to make the fourth-order result independently inspectable without requiring the reader to reconstruct the mathematics from code.

A.1 Even-Power Reduction

For a positive even null order $n = 2p$, the nulling envelope can be written as a finite cosine series:

$$\sin^{2p} x = \frac{1}{4^p}\left[\binom{2p}{p} + 2\sum_{j=1}^{p}(-1)^j\binom{2p}{p-j}\cos(2jx)\right]. \tag{A.1}$$

One destructive output contains this envelope multiplied by an imaging factor $\cos^2(\psi \pm \pi/4)$. For an axisymmetric source, the complementary imaging phase contributes one half after full azimuthal integration. Let a sky point at angular radius ϑ have projected nulling coordinate $\alpha = \vartheta\cos\varphi$. Substituting $x = \pi b\vartheta\cos\varphi/\lambda$ into Eq. A.1 and using the NIST Digital Library of Mathematical Functions integral representation [15],

$$\frac{1}{2\pi}\int_0^{2\pi}\cos(z\cos\varphi)\,\mathrm{d}\varphi = J_0(z) \tag{A.2}$$

gives Eq. 4.2. The reduction is exact for the proxy response and any circularly symmetric surface-brightness profile. It is not an approximation based on rapid rotation.

For the two orders used in the thesis,

$$\langle T_2\rangle_\varphi = \frac{1}{4}\left[1 - J_0\left(\frac{2\pi b\vartheta}{\lambda}\right)\right], \tag{A.3}$$

$$\langle T_4\rangle_\varphi = \frac{1}{16}\left[3 - 4J_0\left(\frac{2\pi b\vartheta}{\lambda}\right) + J_0\left(\frac{4\pi b\vartheta}{\lambda}\right)\right]. \tag{A.4}$$

At large phase, the oscillatory Bessel terms approach zero. The corresponding far-field averages are therefore $1/4$ and $3/16$. This is the origin of the throughput caveat in the main text.

A.2 Uniform-Disk Average and Small-Star Limit

For a uniform disk of angular radius R, area averaging a radial Bessel term gives, using the Bessel derivative identity [15],

$$\frac{2}{R^2}\int_0^R \vartheta J_0(a\vartheta)\,\mathrm{d}\vartheta = \frac{2J_1(aR)}{aR}. \tag{A.5}$$

The right-hand side is evaluated as one when $aR = 0$. With $x = 2\pi bR/\lambda$, the disk-averaged responses become

$$\langle T_2\rangle_{\text{disk}} = \frac{1}{4}\left[1 - \frac{2J_1(x)}{x}\right], \tag{A.6}$$

$$\langle T_4\rangle_{\text{disk}} = \frac{1}{16}\left[3 - \frac{8J_1(x)}{x} + \frac{2J_1(2x)}{2x}\right]. \tag{A.7}$$

The second-order result is not new: Eq. A.5 applied to the double-Bracewell response reproduces the uniform-disk stellar leakage given by Lay [10]. What the derivation above adds is the extension to arbitrary even null order through the general coefficient sum of Eq. 4.2, and its reuse inside the one-dimensional radial quadrature for tapered and extended sources in Section 4.2, neither of which follows from the second-order case alone. Using the standard Bessel power series [15], $J_1(x) = x/2 - x^3/16 + x^5/384 + \mathcal{O}(x^7)$, which yields

$$\langle T_2\rangle_{\text{disk}} = \frac{x^2}{32} - \frac{x^4}{768} + \mathcal{O}(x^6), \tag{A.8}$$

$$\langle T_4\rangle_{\text{disk}} = \frac{x^4}{256} + \mathcal{O}(x^6). \tag{A.9}$$

Consequently,

$$\frac{\langle T_4\rangle_{\mathrm{disk}}}{\langle T_2\rangle_{\mathrm{disk}}} = \frac{x^2}{8} + \mathcal{O}(x^4). \tag{A.10}$$

This scaling predicts the near-disappearance of geometric stellar leakage before any mission scheduling is performed. Direct positive quadrature is retained as a numerical safeguard where subtraction of nearly equal Bessel terms could lose precision.

A.3 Radial Profiles and Wavelength Bins

For a circular source with surface brightness $I_\lambda(\vartheta)$, the transmitted spectral photon rate is proportional to

$$2\pi \int_0^{R_{\max}} I_\lambda(\vartheta)\, \langle T_n\rangle_\varphi (\vartheta)\vartheta \,\mathrm{d}\vartheta. \tag{A.11}$$

This single radial integral replaces a two-dimensional pixel sum. The stellar disk terminates at its angular radius; the exozodiacal calculation uses the adopted smooth radial brightness profile and field-of-view taper. A source lacking circular symmetry cannot be reduced in this way.

For wavelength-bin edges λ_- and λ_+, the calculation changes variables to $u = 1/\lambda$:

$$\int_{\lambda_-}^{\lambda_+} f(\lambda)\,\mathrm{d}\lambda = \int_{1/\lambda_+}^{1/\lambda_-} \frac{f(1/u)}{u^2}\,\mathrm{d}u. \tag{A.12}$$

Eight-point Gauss–Legendre quadrature is then applied over the finite u interval. Because the interferometric phase is proportional to u, this samples chromatic fringes more uniformly than the same number of points placed directly in wavelength.

A.4 Local-Zodiacal Solid Angle

For a locally uniform foreground with spectral radiance I_λ, the photon rate is proportional to the accepted solid angle. In the small-angle approximation, a circular field with angular radius R_{FoV} has

$$\Omega_{\mathrm{circ}} = \pi R_{\mathrm{FoV}}^2. \tag{A.13}$$

In its tapered configuration the removed grid implementation averaged the transmission over the whole square evaluation grid, of area $\Omega_{\mathrm{square}} = 4R_{\mathrm{FoV}}^2$, and then multiplied that average by the circular solid angle above. Its untapered configuration instead averaged over a circular mask, and so paired the two consistently; the disagreement between the two branches on the same physical configuration is what identifies the tapered one as defective. For a foreground and a transmission that are both uniform over the domain, the mismatch is the ratio of the domains,

$$\frac{\Omega_{\text{circ}}}{\Omega_{\text{square}}} = \frac{\pi}{4}, \tag{A.14}$$

so the corrected rate is the former rate multiplied by $4/\pi$. The transmission actually used is neither uniform nor confined to the circle, and the corner flux admitted by the square domain makes the realized ratio slightly smaller; measured against the removed grid and extrapolated in resolution it converges to about 1.267. The value $4/\pi$ is therefore the uniform-domain limit of the correction rather than its exact magnitude. The current analytic calculation uses the circular solid angle directly, consistent with the continuous field-of-view convention in the LIFE source model [2].

B

Appendix B

Reproducibility and Statistical Interpretation

Operating-point records and interpretation of the sampled mission-time percentile

B.1 Operating Point and Search Products

Unless an axis is explicitly varied, every reported contour uses limiting K-band magnitude 7, implemented field of regard $|\beta_{\mathrm{ecl}}| \leq 65°$, and 12 h slew time. The flat, short-weighted, and long-weighted families are evaluated independently. A ramp amplitude always denotes its non-zero endpoint, and its other endpoint is constrained to zero by definition.

The completed products are sufficient to reproduce the reported contour endpoints and figures from the saved AMS outputs. They are not sufficient to recompute a different population quantile or a confidence interval in years because individual universe times were not retained for every evaluated amplitude. Reproducibility of future studies would improve by storing, for every search evaluation:

- the complete per-universe detection, orbit, characterization, and slew times;
- the selected target identifiers and one-hour SNRs;

- the lower and upper amplitude bracket and the stopping reason;
- the exact catalog, source-model, and scheduler configuration;
- the proxy null order and whether throughput was normalized.

B.2 Data and Source Provenance

The reformulation of the astrophysical background evaluation described in Section 4.2 was documented in a standalone technical brief covering its derivation, its validation boundaries, and the local-zodiacal normalization correction, and was reviewed and accepted by the thesis supervisor. The code state carrying that reformulation has since been placed under version control; the source hash below remains the authoritative identifier for the production campaign, because the working tree was not clean when those calculations were run.

Table B.1 records the exact local inputs and result closure used for the final thesis. SHA-256 hashes make later identity checks possible even though no immutable public archive or complete environment lock was created during the production campaign. The source hash covers the production-code closure used by the completed calculations; the result hash covers the 52 files in the complete `TSE Linear` result tree from which the thesis figures and values were taken.

Artifact	Size or count	SHA-256
Hab2Min catalog	126 552 381 bytes	`8a43af800ed5bb7540a0656506057dbb` `4f9dabc6be553fcf510f767ccd751aee`
Hab2Max catalog	190 077 716 bytes	`c372e4f665856ceb2ea7dc4c0e059b84` `47a119586e75e74e98001c3425da9799`
Production source closure	commit `2fd5142`, plus recorded working tree	`00751d7dca7c095819401bd95c8ccfb6` `4a5778ccd39469c9935eb29673b12b90`
Complete final result tree	52 files; 1 895 071 bytes	`61889403be458b5ac65bc7c63bde4eeb` `f260a6251d5ed4bca3e0c2dd1d3181cc`
Validation scripts	recorded script closure	`040dd0bca26ba14c17a2b8cbed2745e0` `fa444d0c8956c851e388e442fc1897b1`

Table B.1: Identity record for the catalogs, production source, completed result tree, and validation scripts used in this work. The repository was not clean at production time; the source-tree hash, rather than the commit identifier alone, defines the recorded code state.

The retained `TSE Linear` products contain rendered PDFs and text summaries, not raw per-evaluation grids, logs, or per-universe time vectors. The order-four Hab2Min 8-year ramp values were recovered from the complete vector result PDFs and cross-checked by reproducing the corresponding 7.5-year values before rounding to the precision used in

Table 5.1. Consequently, the final figures and tabulated endpoints are traceable, but the production campaign cannot be rerun bit-for-bit from this record alone. A practical future release should archive the catalogs or their immutable source identifiers, the full code closure, an environment lock, the run matrix, raw grids, scheduler logs, and per-universe outputs.

B.3 Independent Reproduction of the Final Amplitudes

The amplitudes of Table 5.1 were produced during a campaign in which the working tree was not clean, and the raw per-evaluation products were not retained. To establish that they are nevertheless reproducible, the complete endpoint search was rerun from a committed code state at the identical operating point, using a single dedicated script whose output is the machine-written table below. The rerun additionally records the mission time that each reported amplitude actually delivers, which the original campaign did not retain.

That last quantity matters for interpretation. The search stops once the evaluated mission time lies within 0.05 yr of the target, and it returns its last accepted amplitude, so a reported value belongs to the achieved time rather than to the nominal target. Across the twenty-four reruns the residuals span -0.045 to $+0.045$ yr, filling the stopping tolerance almost exactly. Converting a residual into an amplitude uncertainty requires the local slope of amplitude against mission time, which the secant between each catalog's two targets estimates as roughly 6×10^2 to $1.5 \times 10^3\,\mathrm{ph\,s^{-1}\,\mu m^{-1}}$ per year. The stopping tolerance therefore corresponds to an amplitude uncertainty of order 20 to 80 $\mathrm{ph\,s^{-1}\,\mu m^{-1}}$, depending on the row. Because the response is not linear in mission time, these figures indicate the scale of the search uncertainty rather than a calibrated interval, and they are the reason no reported amplitude should be read to better than its leading two digits.

Of the twenty-four amplitudes in Table 5.1, twenty reproduce to the precision reported there and four differ by a single rounding step: the Hab2Max six-year order-four short-weighted value, the Hab2Min seven-and-a-half-year flat values at both orders, and the Hab2Min seven-and-a-half-year order-two long-weighted value. All four differences are smaller than the search uncertainty estimated above, and no difference exceeds one step. The Hab2Min eight-year rows reproduce throughout; those values had previously been recovered from vector plots rather than from retained numerical output, and are now re-derived directly from the code.

The reproduction also exposes an inconsistency in the reporting convention rather than in the calculation. Table 5.1 rounds most amplitudes to the nearest ten, but the smallest entry is given as 65, which is neither the nearest ten nor two significant figures of the reproduced 62.2. Small amplitudes are where the fixed time tolerance translates into the largest relative uncertainty, so they are also where a rounding convention matters least and misleads most; the values in Table B.2 should be preferred where the two disagree.

Catalog	Target [yr]	Flat	Short-weighted Amplitude [ph s^{-1} μm^{-1}]	Long-weighted	Achieved time [yr]
Null order two					
Hab2Max	5.5	142.4	298.3	210.5	5.455–5.541
Hab2Max	6.0	610.3	1303.1	1089.3	5.962–6.045
Hab2Min	7.5	62.2	197.9	156.7	7.484–7.528
Hab2Min	8.0	462.8	973.9	810.1	7.992–8.037
Null order four					
Hab2Max	5.5	641.2	1080.1	1474.0	5.456–5.519
Hab2Max	6.0	943.9	1840.7	2240.5	6.011–6.034
Hab2Min	7.5	596.5	1001.2	1263.5	7.490–7.531
Hab2Min	8.0	786.2	1458.3	1864.1	7.977–8.037

Table B.2: Reproduction of the final amplitudes of Table 5.1 from a committed code state at the identical operating point. Values are the raw search results, unrounded. The final column gives the range of mission times actually delivered by the three amplitudes in that row; the search stops within 0.05 yr of the target, so a reported amplitude belongs to its achieved time rather than to the nominal target.

B.4 Rank Uncertainty of the 90th Percentile

Let $q_{0.9}$ be the population 90th percentile and let $X_{(k)}$ denote the kth ordered value in a sample of 100 independent population draws. The number of samples below $q_{0.9}$ follows a binomial distribution with $n = 100$ and success probability 0.9. Therefore the coverage probability of the interval $[X_{(84)}, X_{(96)}]$ is obtained from the corresponding binomial tail sum and is approximately 0.9557.

This interval quantifies rank uncertainty under independent sampling from the adopted population model. It does not include uncertainty in planet occurrence rates, stellar properties, dust levels, instrument parameters, or scheduler design. The universes also share the same real-star scaffold. The rank statement is therefore useful but narrower than a complete physical uncertainty analysis.

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